

Sri Lanka Institute of Information Technology



Assignment 02 Report

Data Warehousing and Business Intelligence

IT3021

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Step 1: Data source for the Assignment 2

PerishableDW Data Warehouse was designed and implemented in Assignment 01 using SQL Server and the same data warehouse used for the Assignment 02 as the data source. This PerishableDW data warehouse follows a star schema architecture consisting of a central fact table and multiple dimension tables connected to the fact table. It stores processed and integrated data related to perishable goods transactions, enabling analytical process and business intelligence reporting.

Grain of the Fact Table

The grain of the fact table defined as one record per product transaction per store per day meaning, each row represents a unique transaction event capturing sales, waste, pricing details for a specific product at a specific store on a given date.

Fact Table:

Measures in the fact table like revenue, units_sold, profit are primarily additive that can be aggregated across all dimensions to support efficient summarization and OLAP analysis.

Fact_Transaction – store transactional data

- Units_sold
- Units_wasted
- Revenue
- Profit
- Waste_cost
- Base_price
- Cost_price
- Selling_price
- Expiration_date
- Product_key (FK)
- Store_key (FK)
- Supplier_key (FK)
- Storage_key (FK)
- Date_key (FK)

Dimension Tables:

Dim_Product – product related information (SCD Type 2)

- Product_key (PK)
- Product_id
- Product_name
- Category
- Spoilage_sensitivity
- Start_date
- End_date
- Is_current

Dim_Store – store and regional information

- Store_key (PK)
- Store_id
- region

Dim_Supplier – supplier performance and details (SCD Type 2)

- Supplier_key(PK)
- Supplier_id
- Supplier_score
- Start_date
- End_date
- Is_current

Dim_Storage – storage and handling conditions

- Storage_key (PK)
- Storage_temp
- Temp_deviation
- Handling_score
- Packaging_score

Dim_Date – supports time-based information

- Date_key(PK)
- Full_date
- Day
- Month
- Year
- Day_of_week
- Is_weekend
- Quarter
- Week_of_year

Surrogate keys are used to connect the fact table to all dimension tables, enabling efficient joins and fast query performance. This data warehouse supports analytical queries, OLAP operations, and reporting tools such as SSAS, Excel and Power BI. Multiple sources like CSV files, Excel files and SQL Server were used to originate and integrated the dataset transactional data using SSIS ETL process. Overall, the data warehouse serves as a centralized repository for analysing sales, revenue, spoilage and operational efficiency in the perishable domain.

SCD Type 2

Dim_Product and Dim_Supplier dimension tables are implemented as Slowly Changing Dimensions (SCD Type 2) to preserve historical data allowing track changes over time without losing previous records which essential for accurate historical analysis.

ETL Process (Extract, Transform, Load)

The data warehouse was implemented using SSIS ETL process which extract data from multiple sources like Excel files, CSV files and SQL Server databases transforming it into a consistent format and loaded into dimensions and a fact table.

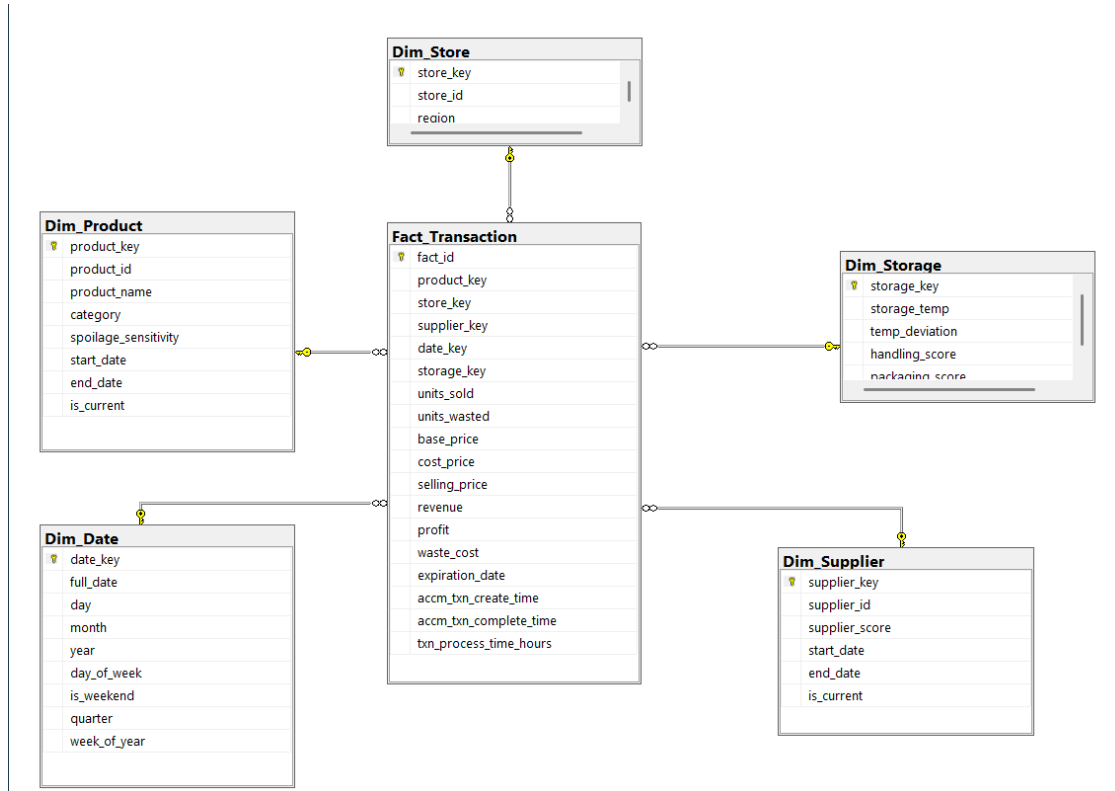


Figure 1: Star Schema of PerishableDW

The above relational model diagram illustrates the star schema design of the PerishableDW data warehouse. The Fact_Transaction fact table is centrally positioned and connected to dimension tables including Dim_Product, Dim_Store, Dim_Supplier, Dim_Storage and Dim_Date through foreign key relationships.

This data warehouse supports reliable, efficient querying, analytical processing and optimized foundation for building OLAP cubes and performing multidimensional analysis in SSAS.

Step 2: SSAS Cube implementation

1. Overview

An OLAP cube was developed to enable efficient multidimensional analysis of the PerishableDW data warehouse using SQL Server Analysis Services (SSAS). The cube allows users to perform fast analytical queries and supports business intelligence operations like aggregation, filtering, drill-down analysis.

The purpose of creating cube is to transform relational data into a multidimensional structure that improves query performance and enable advanced analytical capabilities. The cube allows users to quickly analyse key business metrics such as revenue, profit, waste and sales trends across multiple dimensions using aggregation.

2. Create the SSAS Project

A new Analysis Services Multidimensional Project named 'Perishable_SSAS' was created using SQL Server Data Tool (SSDT) as the development environment for building and managing the cubes.

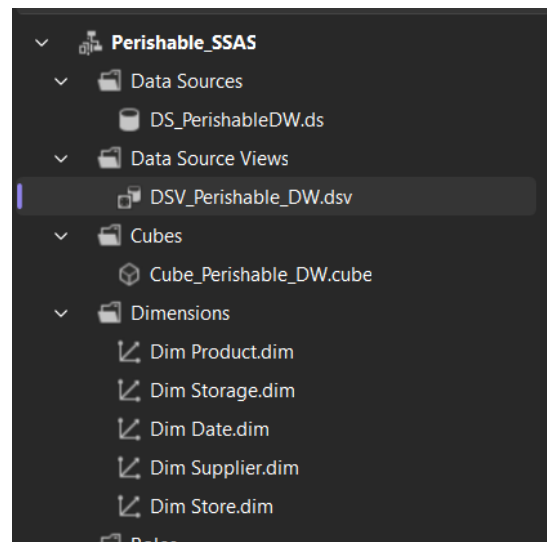


Figure 2: Analysis Services Multidimensional Project structure

3. Configure the Data Source

A data source 'DS_PerishableDW' was created to enable a connection between SSAS and the PerishableDW database to ensure the cube can access the data warehouse tables for processing and analysis.

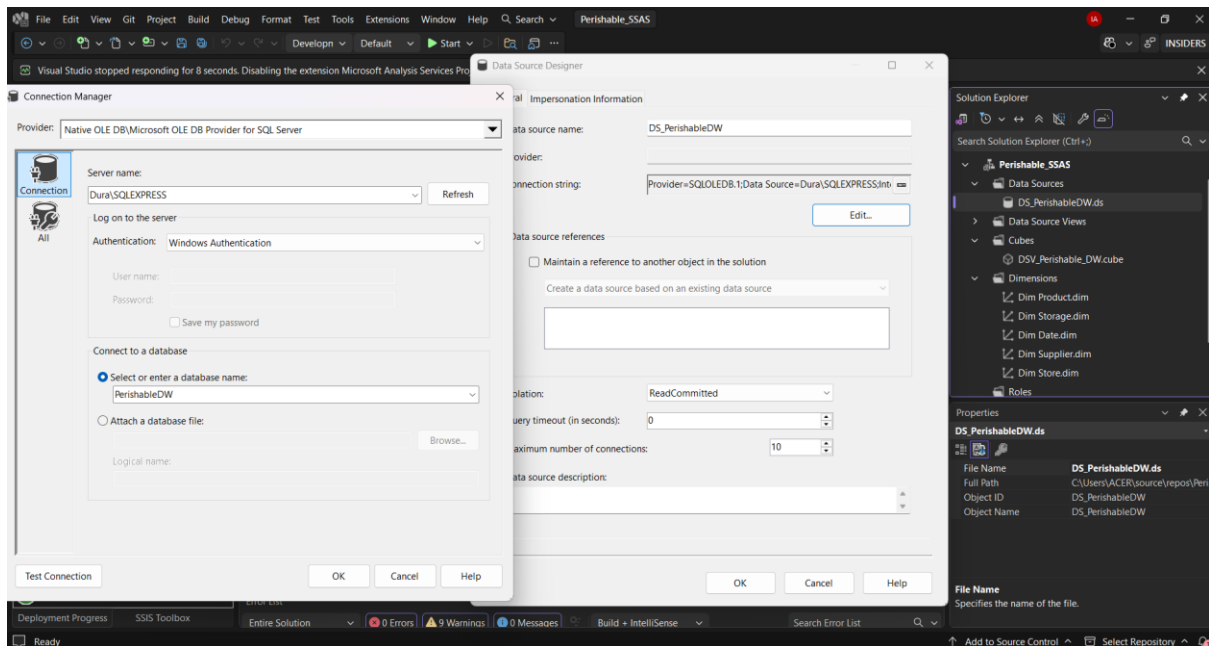


Figure 3: Data Source configuration

4. Create the Data Source View (DSV)

A Data Source View (DSV) created by selecting the Fact_Transaction table along with all the related dimension tables including Dim_Product, Dim_Store, Dim_Supplier, Dim_Storage, Dim_Date. Relationships between fact and dimension tables were manually defined using surrogate keys because the naming conventions of primary and foreign keys were different and this ensures proper linking between measures and dimensions.

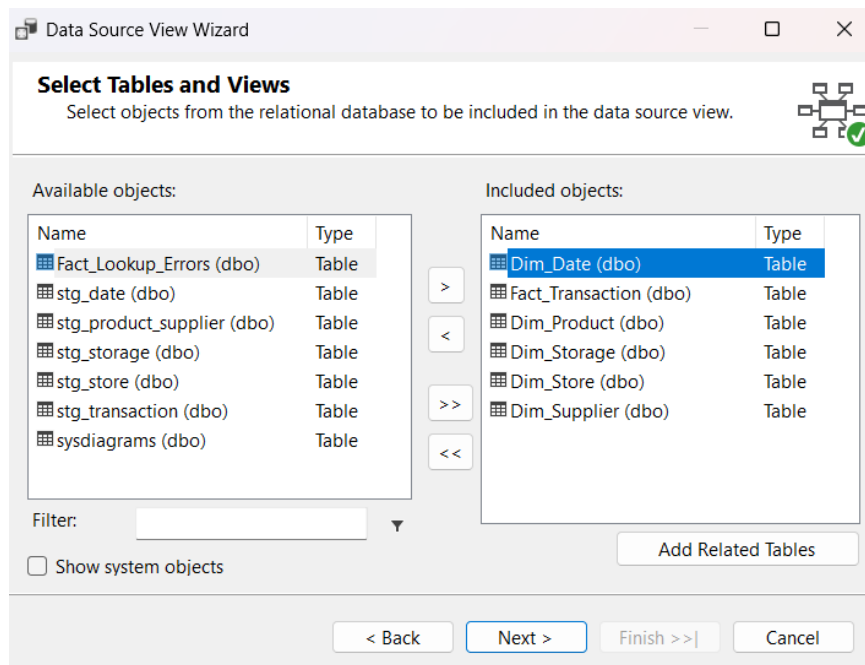


Figure 4: Data Source View Creation

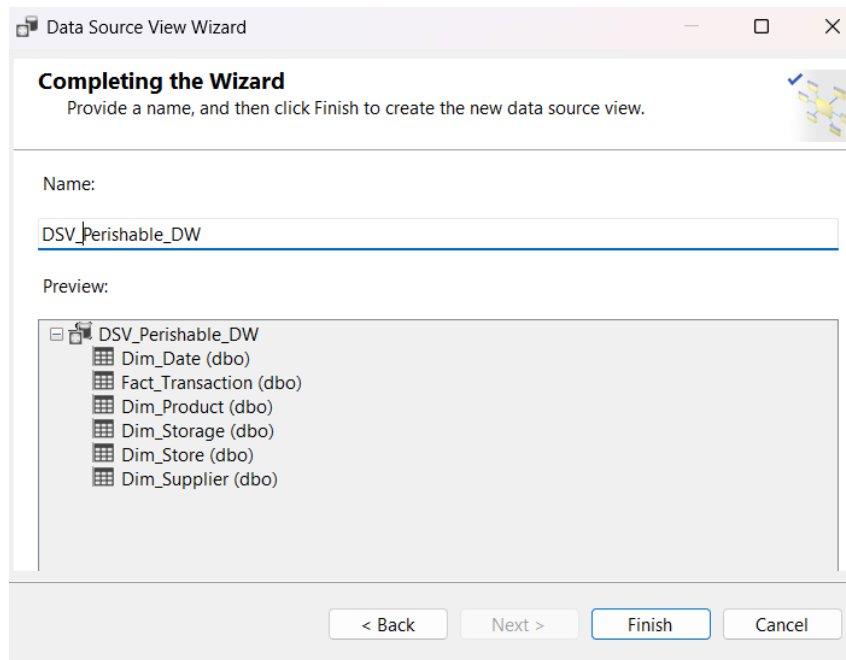


Figure 5: Completion of Data Source View

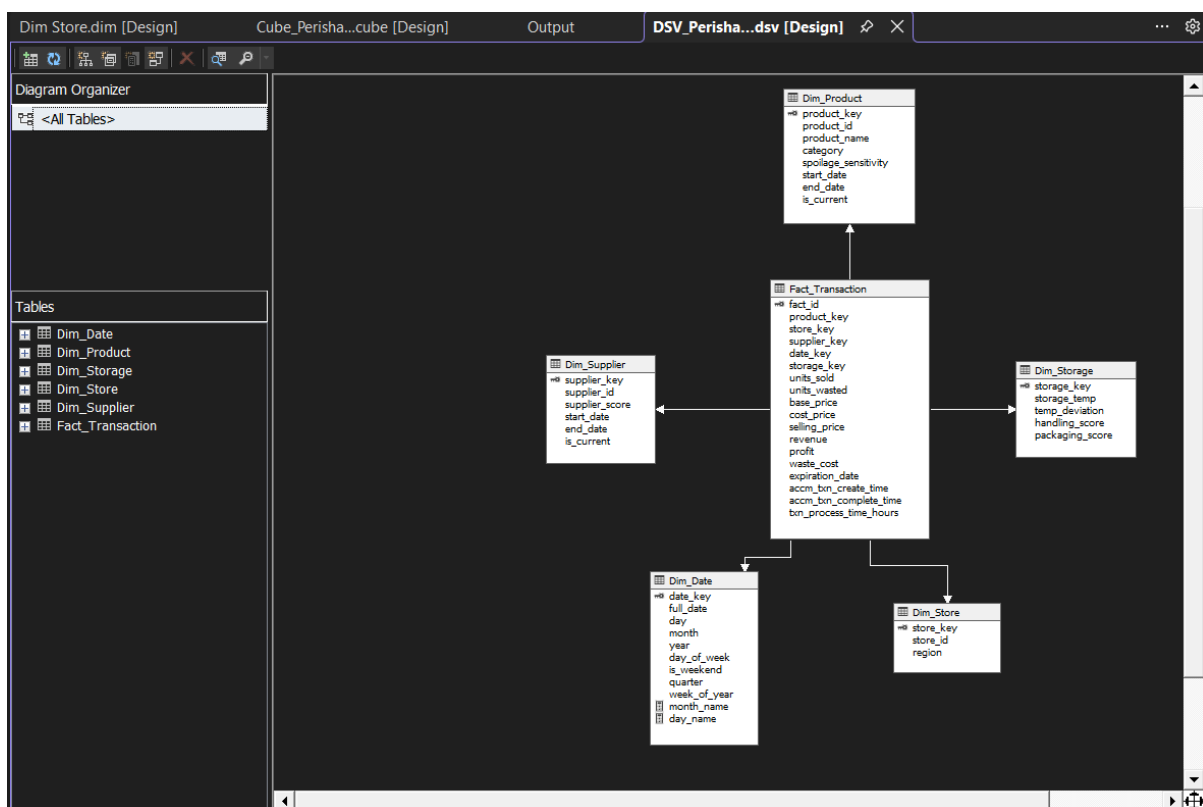


Figure 6: Data Source View

5. Cube Creation

OLAP Cubes known as a method for storing data in a multi-dimensional form that enables users to analyse multidimensional data (facts) interactively (by browsing and querying) from multiple perspectives (dimensions). Pre-calculates most of the queries that are typically time taking to execute over tabular databases such as aggregation, joining and grouping. A data structure optimized for very quick data analysis, especially when optimized for OLAP operations like Roll-up, Drill-down, Slice, Dice and Pivot.

Selected measures in the Fact_Transaction table including revenue, profit, units sold, units_wasted and waste_cost were used to create the cube by using the 'use existing table' option. These measures represent key performance indicators identified for business analysis.

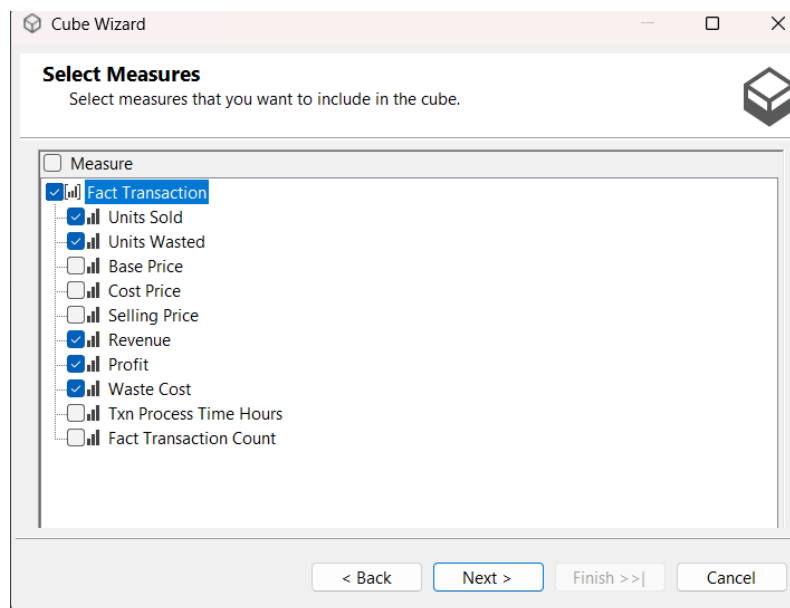


Figure 7: Cube measure selection for fact table

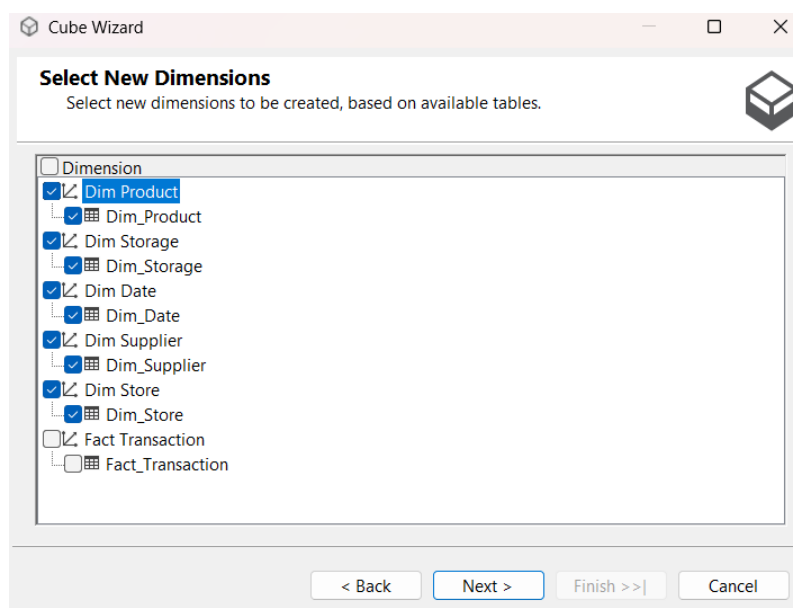


Figure 8: Cube creation for Dimension tables

6. Dimension Configuration

Dimension tables were added to the cube to provide analytical perspectives, and each dimension were configured by adding meaningful attributes. These allows users to analyse measures across different business perspectives. Attribute relationships were defined within dimensions to improve query performance and enhance aggregation efficiency and cube optimization performance.

- Dim_Product – product_name, category
- Dim_Store – store_id, region
- Dim_Supplier – supplier_score, supplier_id
- Dim_Storage – storage_temp, temp_deviation, handling_score, packaging_score
- Dim_Date – year, quarter, month, day, is_weekend, week_of_year, day_of_week

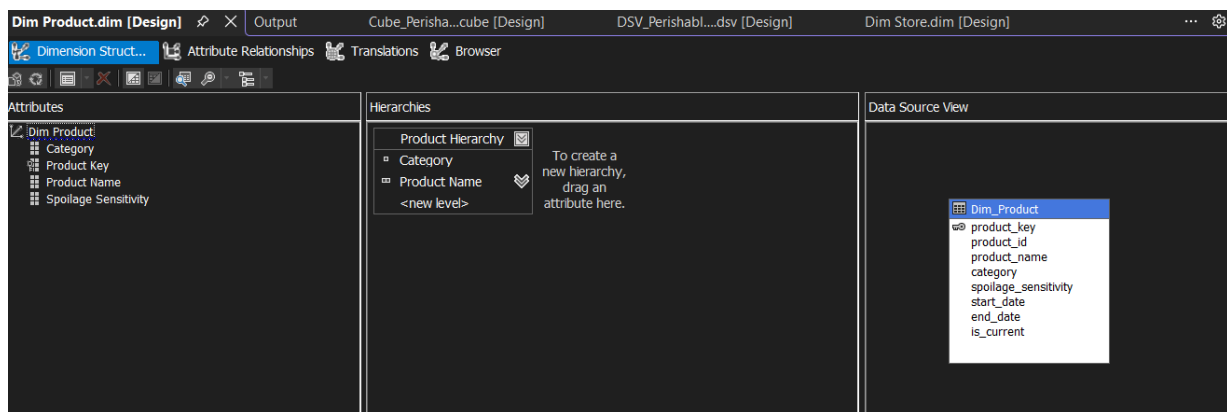


Figure 9: Dimension configuration of Dim_Product

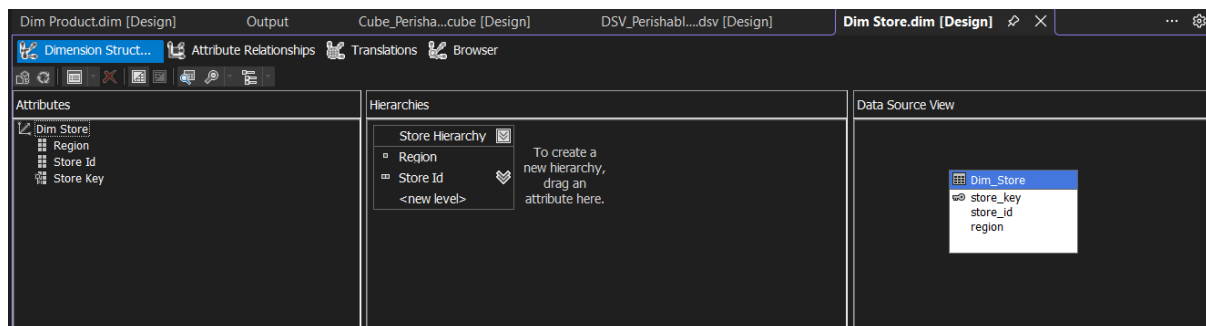


Figure 10: Dimension configuration of Dim_Store

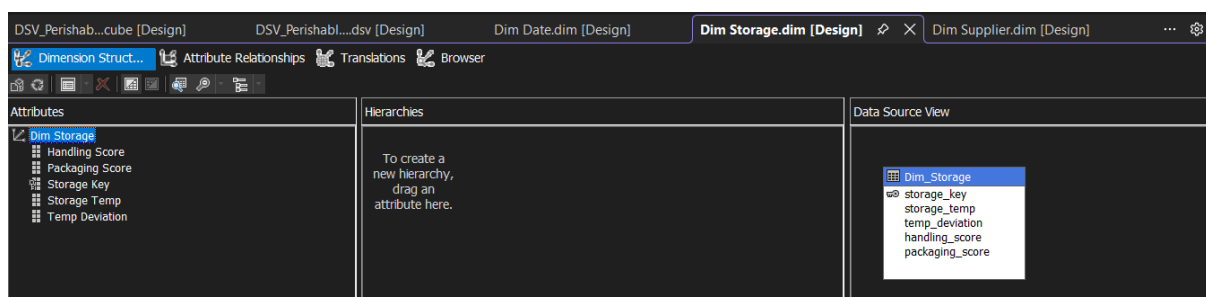


Figure 11: Dimension configuration of Dim_Storage

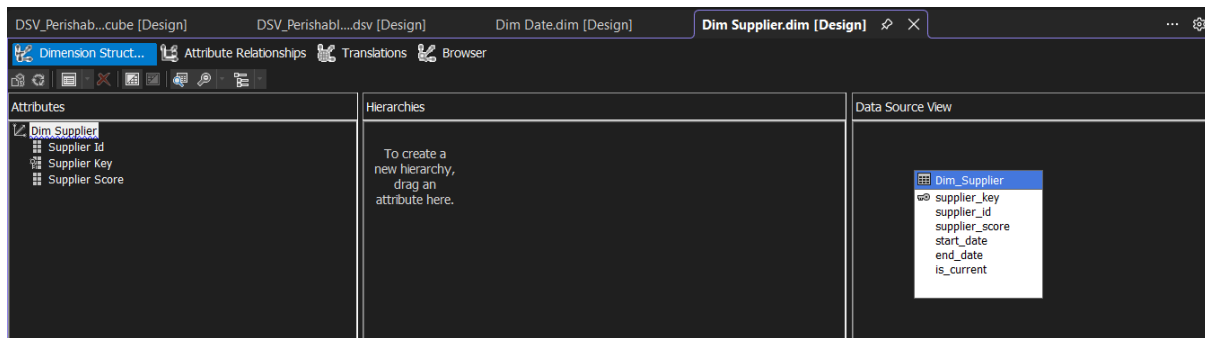


Figure 12: Dimension configuration of Dim_Supplier

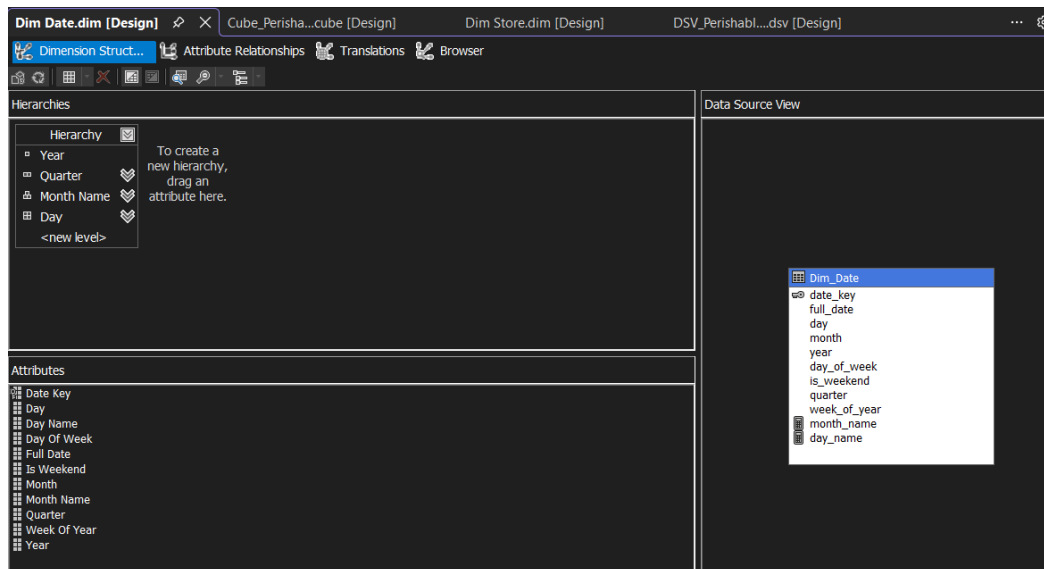


Figure 13: Dimension configuration of Dim_Date

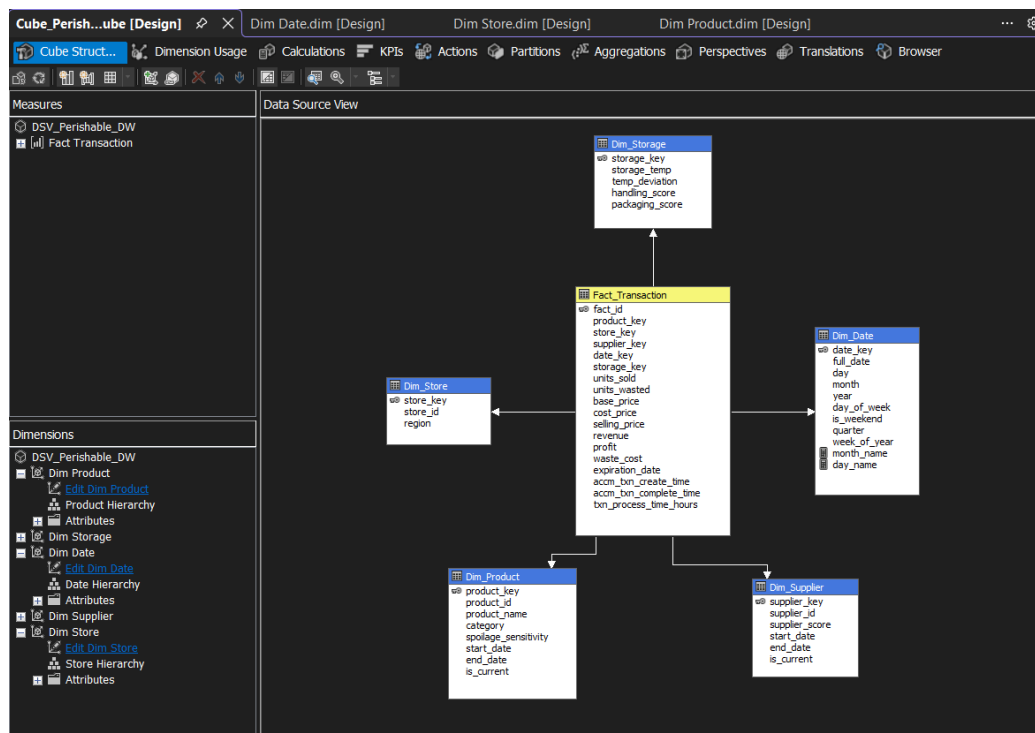


Figure 14: Cube Schema

7. Hierarchy Creation

Attribute relationships in dimensions were defined in a hierarchical manner from lower-level attributes to higher-level attributes like Full-Date, Month, Quarter, Year to ensure proper hierarchical structure. This structure helps to support efficient aggregation and enable optimized drill-down operations within the cube. Hierarchies were created in the Dim_Date, Dim_Product and Dim_Storage dimension tables following up below levels,

7.1 Date Hierarchy – Year → Quarter → Month_Name → Day

The Date Hierarchy was created to support accurate time-based analysis for business intelligence allowing,

- Analyse trends over time
- Identify seasonal patterns and sales fluctuations
- Performs drill-down from year to daily details

Additional attributes like Month_Name and Day_Name were created in the Data Source View using SQL functions (DATENAME) to improve readability and usability of the cube and for the BI purposes. These attributes provide descriptive information instead of numeric values, making the analysis user-friendly.

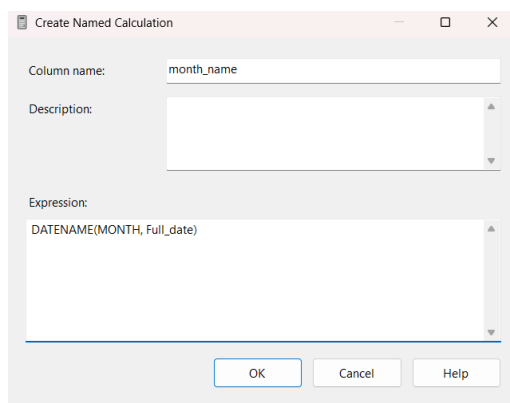


Figure 16: Month_Name Additional Attribute creation

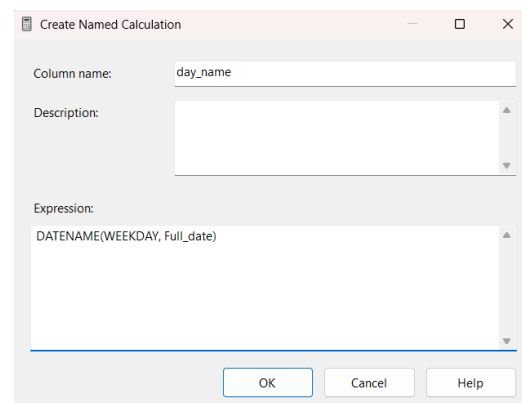


Figure 15: Day_Name Additional Attribute creation

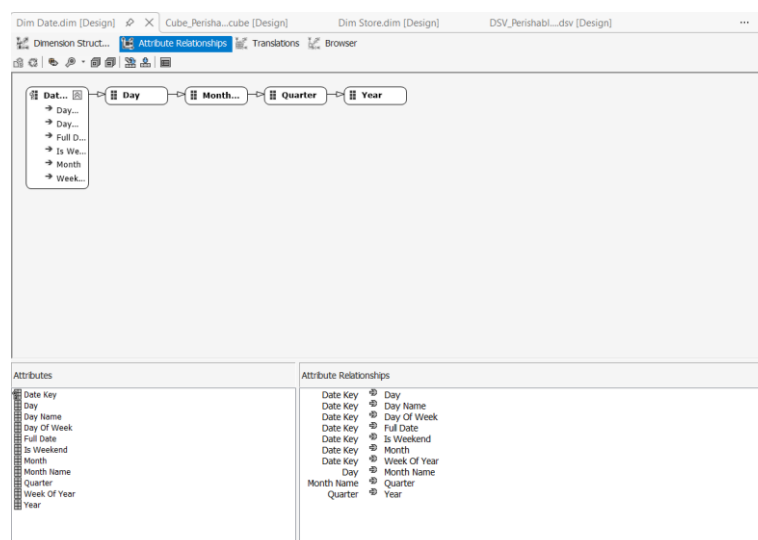


Figure 17: Attribute Relationship of Date Hierarchy

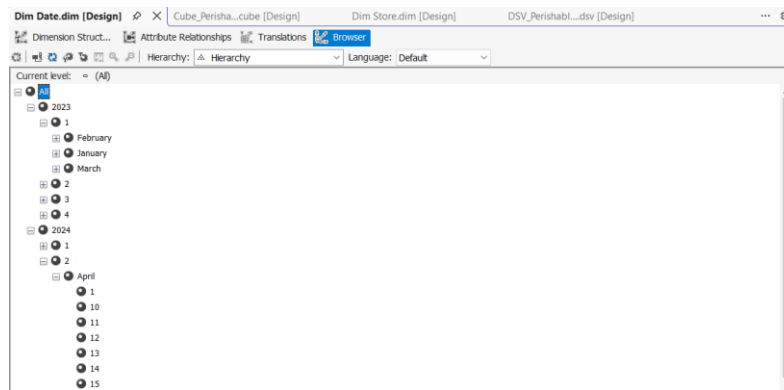


Figure 18: Date Hierarchy Browser

7.2 Product Hierarchy – Category → Product Name

The Product Hierarchy was created to enable product-level performance analysis by grouping products into categories allowing,

- Compare performance across product categories
- Identify best and worst performance of products
- Drill-down to individual product sales

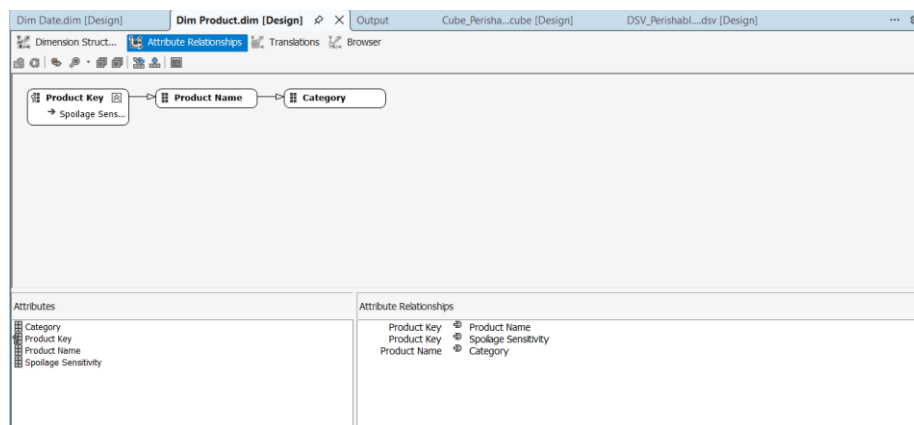


Figure 19: Attribute relationship of Product Hierarchy

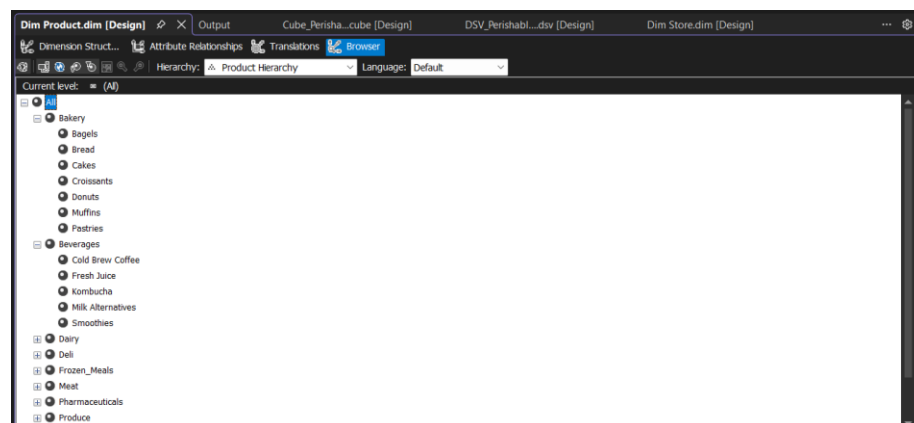


Figure 20: Product Hierarchy Browser

7.3 Storage Hierarchy – Region → Store_id

The Storage Hierarchy was created to supports location- based analysis by organizing stores within the region allowing,

- Analyse regional performance
- Compare stores within the same region
- Identify operational inefficiencies in storage conditions

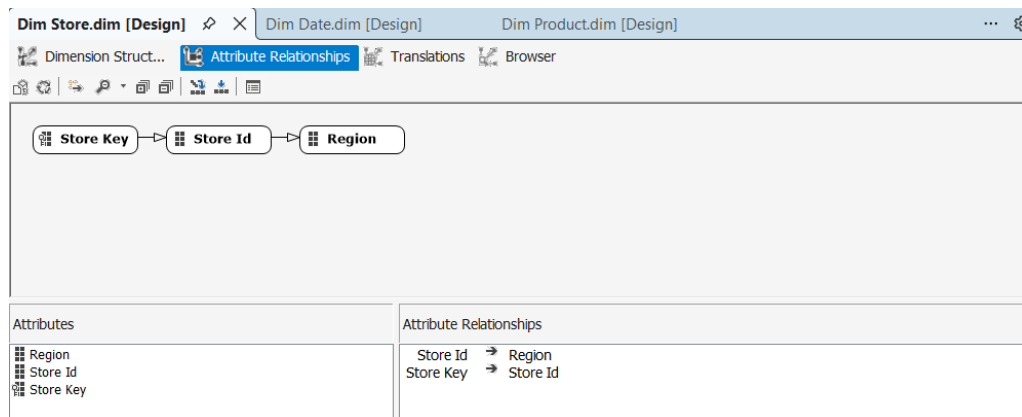


Figure 21: Attribute relationship of Store Hierarchy

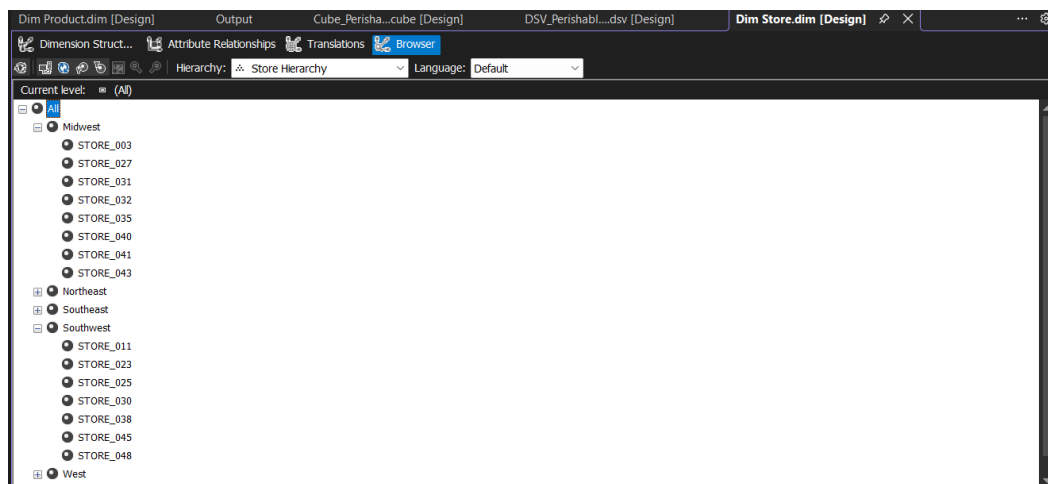


Figure 22: Store Hierarchy Browser

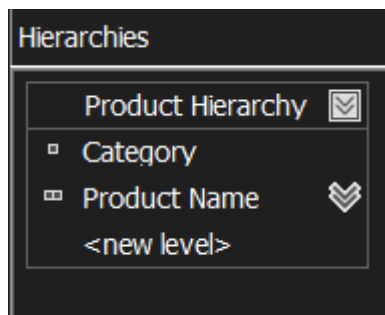


Figure 14: Product Hierarchy

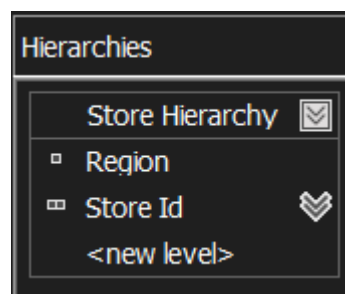


Figure 23: Store Hierarchy

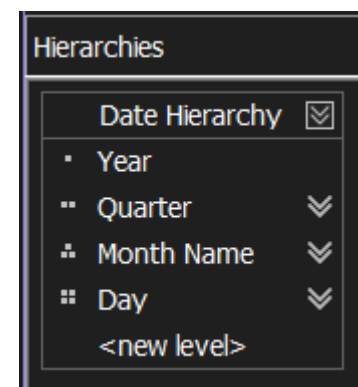


Figure 16: Date Hierarchy

By creating hierarchies can enable drill-down and roll-up operations, allowing users to analysis data in different level of granularity.

7.4 Clarify Hierarchy Types

1. Flexible - used when attributes change frequently over time and not fixed. SSAS assumes parent-child links can change over time.
2. Rigid – used when attributes are not changed over time and fixed. SSAS assumes parent-child link never change.

For SCD Type 2 dimensions most suitable type is flexible and in Perishable_SSAS, all relationship hierarchies like date, product and storage dimensions were used flexible hierarchy type to ensure correct handling of dimension changes during cube processing.

8. Cube Deployment

The cube was first build and deployed to SQL Server Analysis Services making it available for querying and analysis through tools such as Excel, Power BI and SSMS.

The SSAS cube successfully integrated with the fact table and dimension data from PerishableDW and provides a structured environment for analytical purposes and processing. The inclusion of measures, dimensions, and hierarchies ensures that the system supports comprehensive business intelligence analysis.

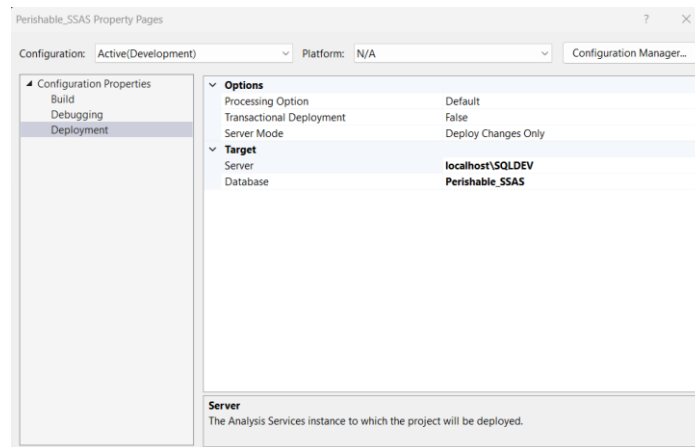


Figure 24: Set up target server and database for deployment

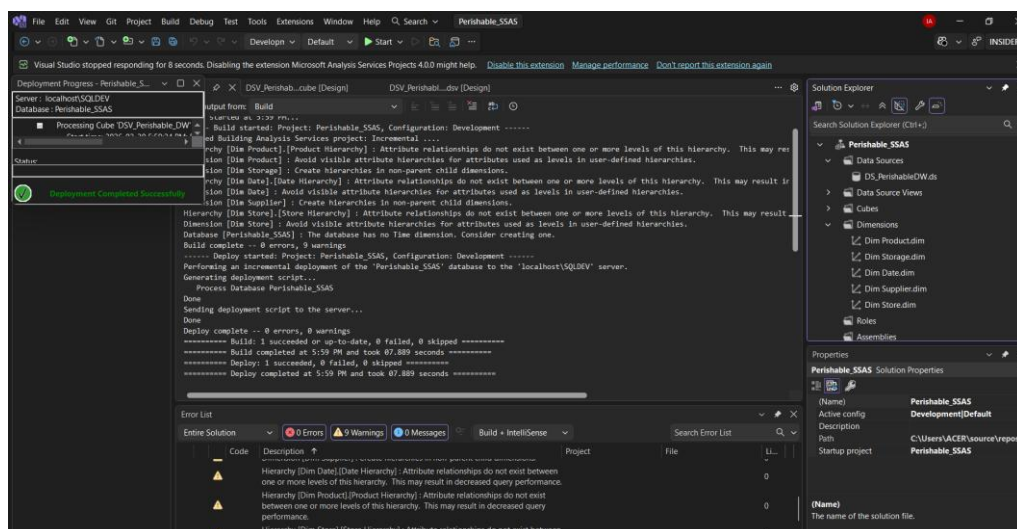


Figure 25: Cube Deployment successful

9. Cube Processing

The cube and its dimensions were processed after deployment to make sure the data from data warehouse were loaded into the multidimensional structure. Processing ensured consistency and correctness that all measures and dimensions are populated and ready for querying.

When processing the cube, SSAS identified duplicate attributes in Product Hierarchy due to 'Day_Name' value has the same day names repeated multiple times within a year. To solve this problem, the Key_Columns for Day_Name were updated to consists Full_Date while keeping the Day_Name as the NameColumn to ensure each attribute key is unique and for successful hierarchy processing.

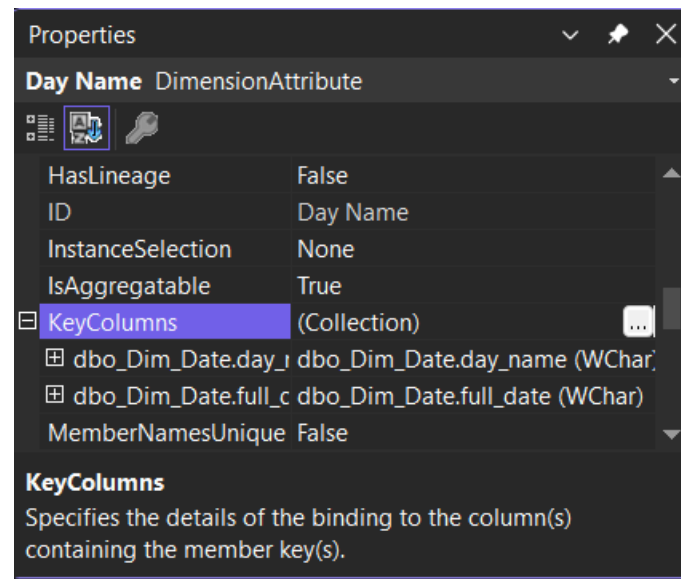


Figure 26: Handle duplicate key issue in Date Hierarchy

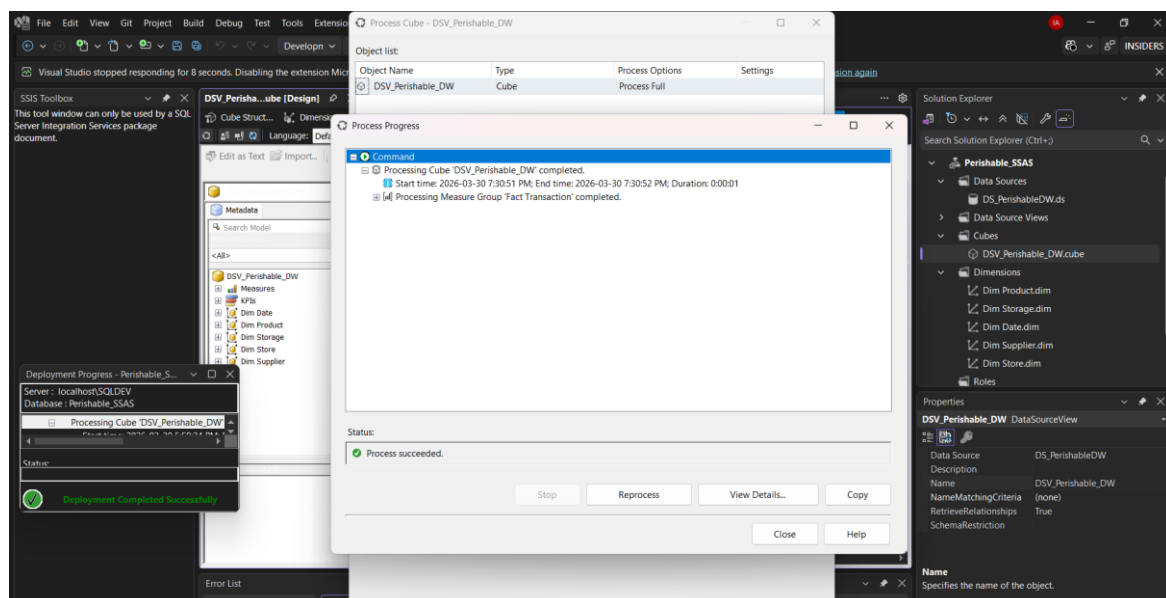


Figure 27: Cube Processing

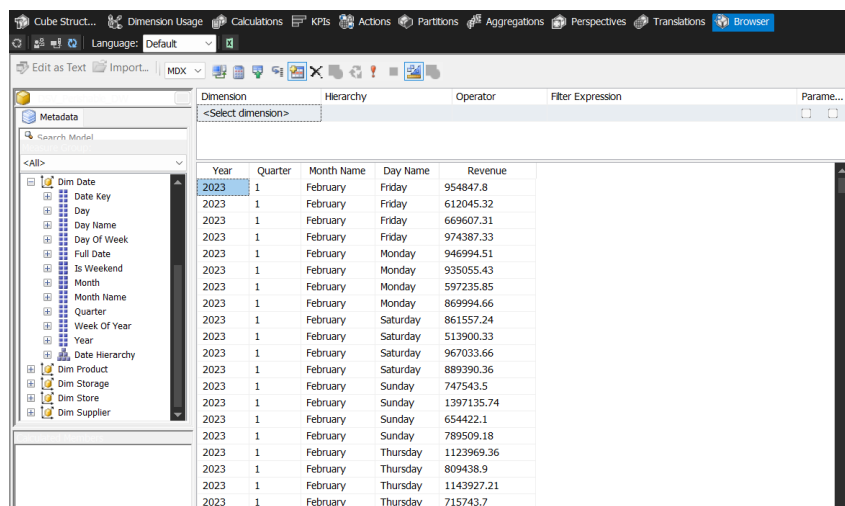
10.Cube Browsing and Testing

The cube was tested using Browser tab in SQL Server Data Tolls after successful deployment and processing. Measures like revenue from Fact_Transaction and dimensions like year from Dim_Date were added into the values areas to analyse data distribution. This confirms the cube successfully returned aggregated output resulting,

- Data source connections were correctly configured
- Relationships between fact and dimensions were valid
- Measures were properly calculated

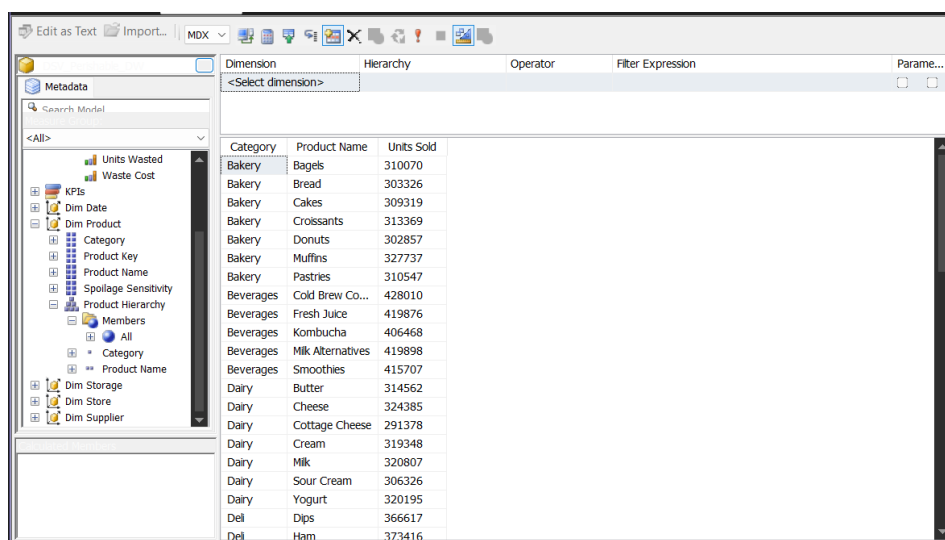
Hierarchy Validation

- Dragging the entire hierarchy displayed all levels expanded, which is expected behavior. This validation ensures that users can explore data at multiple granularities and that OLAP operations function as intended.



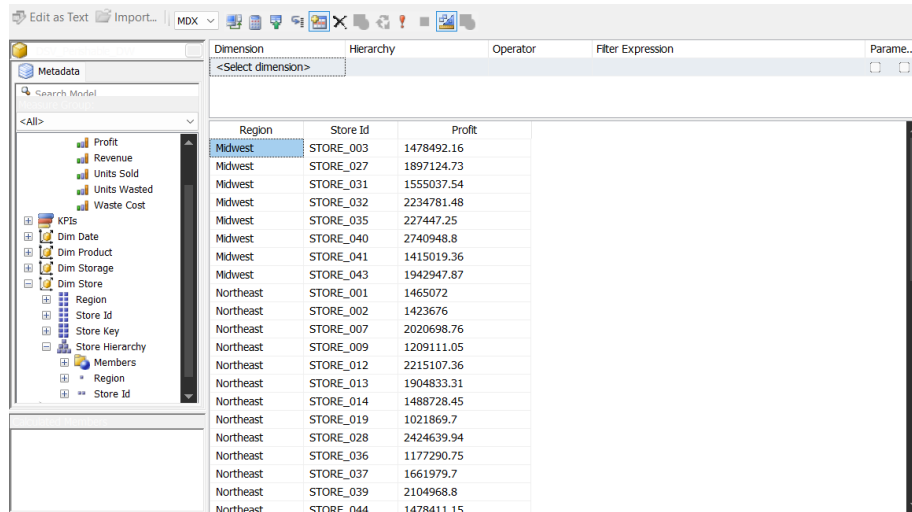
Year	Quarter	Month Name	Day Name	Revenue
2023	1	February	Friday	954847.8
2023	1	February	Friday	612045.32
2023	1	February	Friday	669607.31
2023	1	February	Friday	974387.33
2023	1	February	Monday	946994.51
2023	1	February	Monday	935055.43
2023	1	February	Monday	597235.85
2023	1	February	Monday	869994.66
2023	1	February	Saturday	861557.24
2023	1	February	Saturday	513900.33
2023	1	February	Saturday	967033.66
2023	1	February	Saturday	889390.36
2023	1	February	Sunday	747543.5
2023	1	February	Sunday	1397135.74
2023	1	February	Sunday	654422.1
2023	1	February	Sunday	789509.18
2023	1	February	Thursday	1123969.36
2023	1	February	Thursday	809438.9
2023	1	February	Thursday	1143927.21
2023	1	February	Thursday	715743.7

Figure 28: Date Hierarchy Validation



Category	Product Name	Units Sold
Bakery	Bagels	310070
Bakery	Bread	303326
Bakery	Cakes	309319
Bakery	Croissants	313369
Bakery	Donuts	302857
Bakery	Muffins	327737
Bakery	Pastries	310547
Beverages	Cold Brew Co...	428010
Beverages	Fresh Juice	419876
Beverages	Kombucha	406468
Beverages	Milk Alternatives	419898
Beverages	Smoothies	415707
Dairy	Butter	314562
Dairy	Cheese	324385
Dairy	Cottage Cheese	291378
Dairy	Cream	319348
Dairy	Milk	320807
Dairy	Sour Cream	306326
Dairy	Yogurt	320195
Del	Dips	366617
Del	Ham	373416

Figure 29: Product Hierarchy Validation



The screenshot shows a BI tool interface with a 'Metadata' pane on the left and a main table on the right. The 'Metadata' pane lists various dimensions and measures. The main table displays a hierarchy for the 'Store' dimension, showing 'Region' and 'Store Id' columns. The 'Region' column is expanded, showing 'Midwest' and 'Northeast'. The 'Store Id' column lists specific store IDs, and the 'Profit' column shows the corresponding profit values.

Region	Store Id	Profit
Midwest	STORE_003	1478492.16
Midwest	STORE_027	1897124.73
Midwest	STORE_031	1555037.54
Midwest	STORE_032	2234781.48
Midwest	STORE_035	227447.25
Midwest	STORE_040	2740948.8
Midwest	STORE_041	1415019.36
Midwest	STORE_043	1942947.87
Northeast	STORE_001	1465072
Northeast	STORE_002	1423676
Northeast	STORE_007	2020698.76
Northeast	STORE_009	1209111.05
Northeast	STORE_012	2215107.36
Northeast	STORE_013	1904833.31
Northeast	STORE_014	1488728.45
Northeast	STORE_019	1021869.7
Northeast	STORE_028	2424639.94
Northeast	STORE_036	1177290.75
Northeast	STORE_037	1661979.7
Northeast	STORE_039	2104968.8
Northeast	STORE_044	1478411.15

Figure 30: Store Hierarchy Validation

11. Benefits of creating cube

- Pre-aggregation enables faster query performance
- Simplify data analysis operations without using complex SQL queries.
- Support for OLAP operations such as drill-down, slice, roll-up and dice
- Improve decision making through multidimensional insights

Step 3: Demonstration of OLAP operations

1. Connect Excel to SSAS Cube

Microsoft Excel was connected to the SSAS cube using OLAP functionalities to perform analytical operations. The connection was established by using the Data tab in Excel file and get data from database by clicking analysis services option. The server's name was provided in the window and the Perishable_SSAS cube was selected. The cube was then loaded into Excel as a Pivot Table successfully, enabling interactive data analysis.

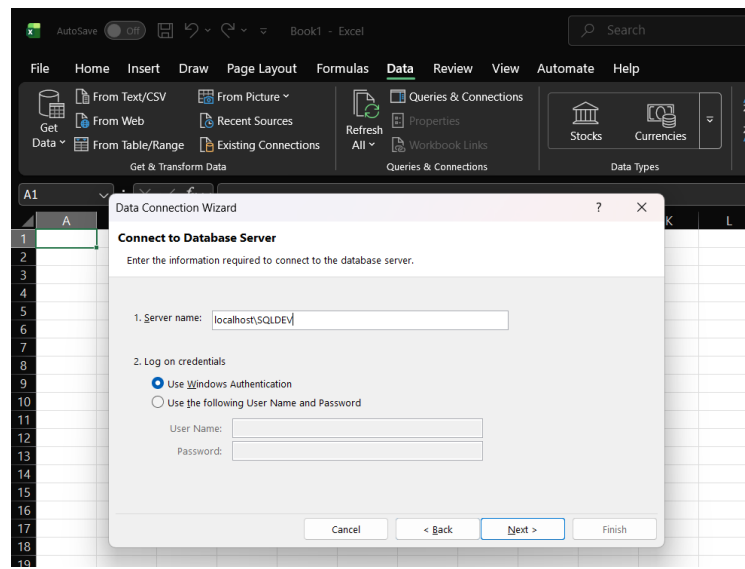


Figure 31: Connecting Excel to SSAS Cube

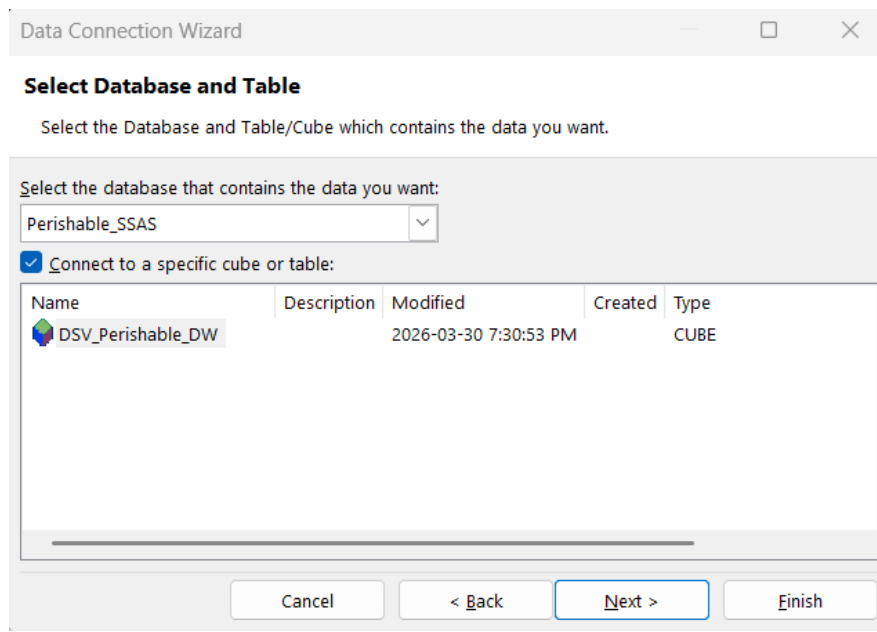


Figure 32: Cube Connection Configuration

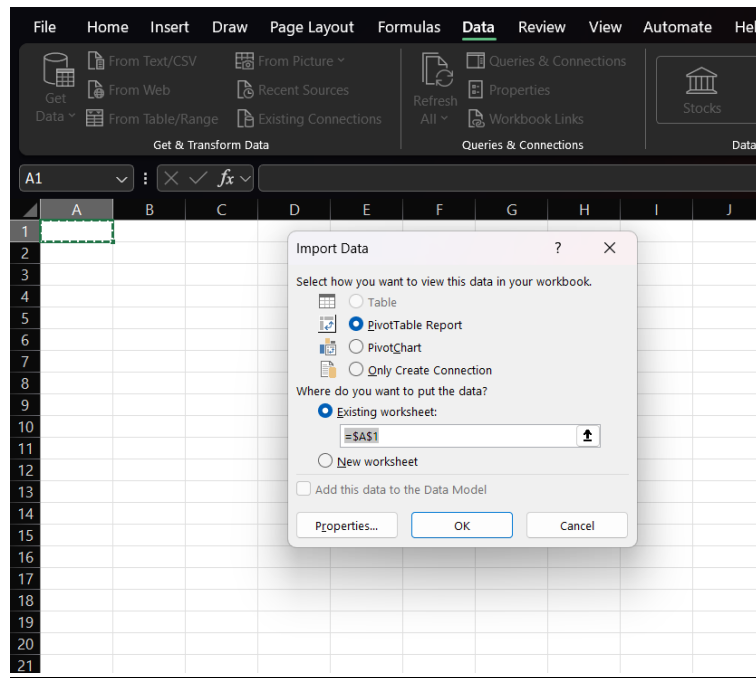


Figure 33: Load into Excel as a Pivot Table

2. Creating Pivot Table for Analysis

A pivot table was created to visualize multidimensional data from the cube. The filed list consists of all measures such as Revenue, Profit, Units Sold and dimensions including Product, Supplier, Storage, Store, Date. The pivot table consists of 4 layout areas as follows:

- Rows – used to display categorical data vertically (ex: Year, Category)
- Columns – used to compare data horizontally (ex: Year, Category)
- Values – used for numerical measurements like Revenue, Profit, Units Sold
- Filters – used to restrict data dynamically (ex: Region, Category)

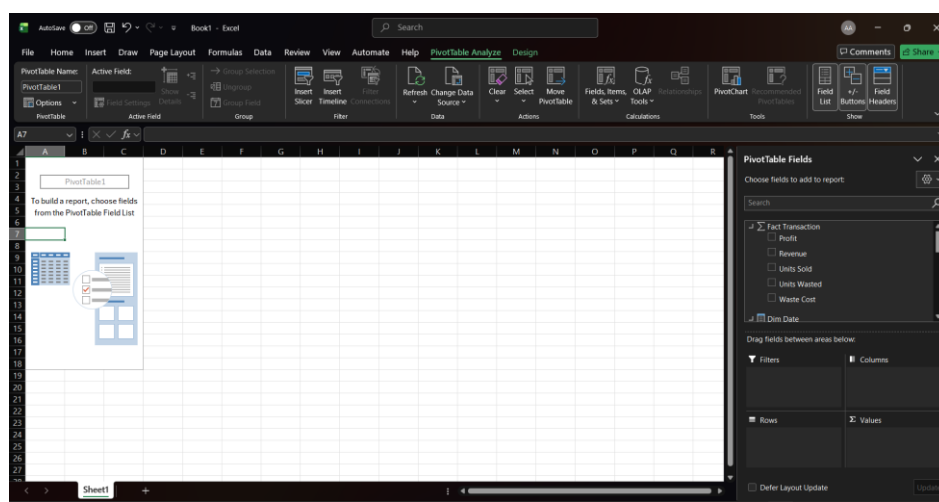


Figure 34: Basic Pivot table

3. Demonstration of OLAP Operations

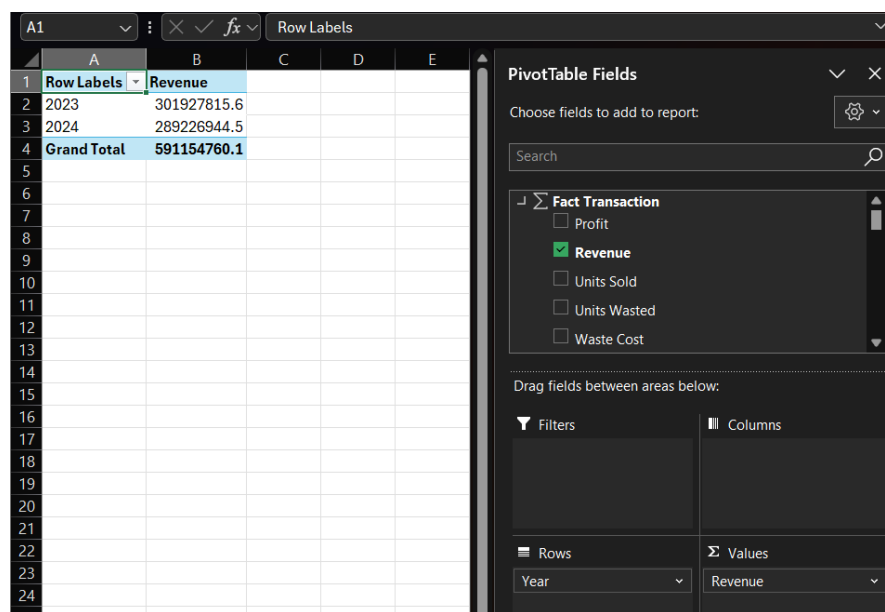
3.1 Roll-Up (Summary View)

The Roll-Up operation demonstrates aggregate data at a high level. Also known as consolidation or aggregation.

Example configuration:

- Rows – Year
- Values – Revenue

The pivot table displayed total revenue for each year (2023-2024) by aggregating data at a high level showing how detailed transactions are aggregated into higher-level summaries for decision making.



Row Labels	Revenue
2023	301927815.6
2024	289226944.5
Grand Total	591154760.1

Figure 35: Roll-Up Operation Revenue by Year

3.2 Drill-Down (Details View)

Drill-Down performs navigation from summary data to more detailed levels like stepping down a hierarchy of a dimension allowing navigation through details. Data is fragmented into smaller parts, and it is the opposite of the rollup process.

Example configuration:

- Rows – Year in Date Hierarchy (expand to Quarter → Month)
- Values – Revenue

Users can expand the Date Hierarchy from Year to quarter, month to view revenue, This enables detailed analysis of trends and helps to identify seasonal variations in sales.

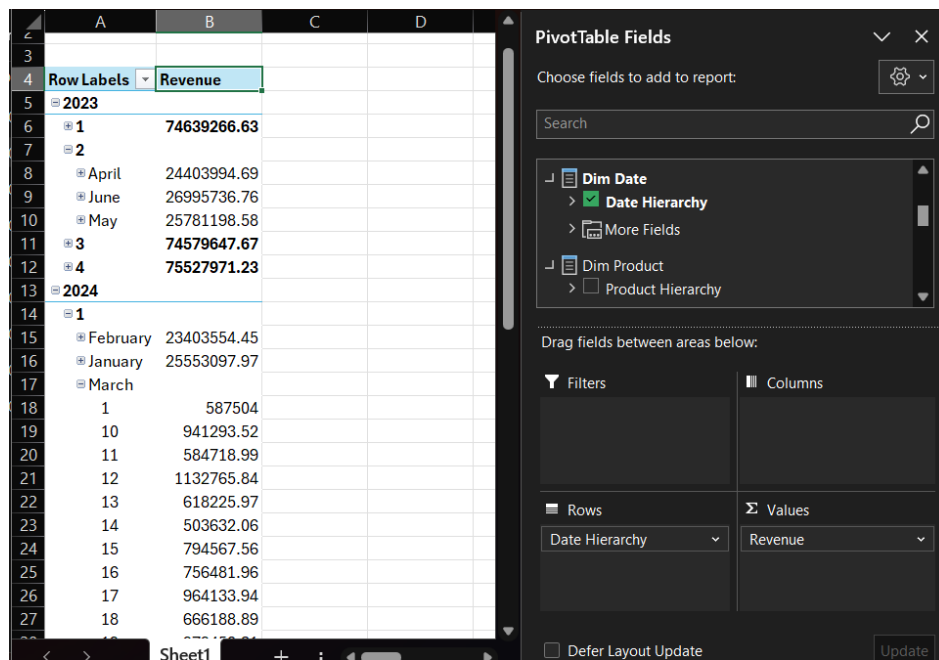


Figure 36: Drill-Down Operation (Revenue by Date Hierarchy)

3.3 Slice (Filter by one Dimension)

Slice is a rectangular subset of a cube, by choosing a single value for one of its dimensions and demonstrated by filtering data using a single dimension.

Example Configuration:

- Rows – Year
- Values – Revenue
- Filter – Region (West)

This results a pivot table of revenue only for the selected region filter by West. This allows to analyse on a specific subset of the data.

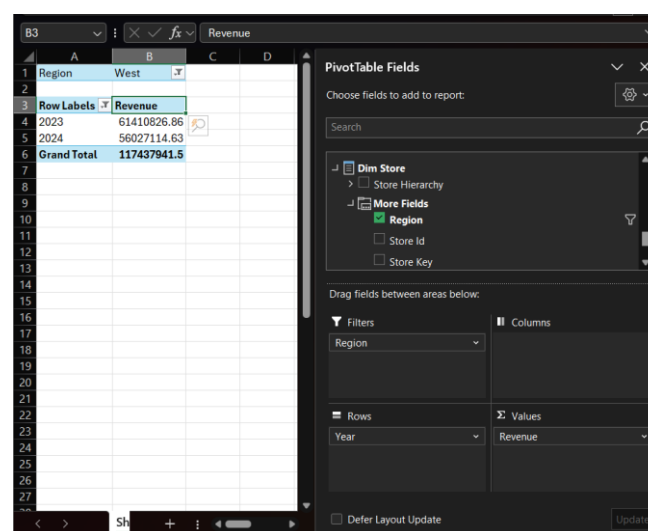


Figure 37: Slice Operation (Year, Revenue filtering by West Region)

3.4 Dice (Filter by Multiple Dimensions)

Dice selects two or more dimensions from a given cube and provides a new sub-cube by selecting specific values on those selected dimensions. Dice operation demonstrates by applying multiple filters across dimensions.

Example configuration:

- Rows – Category / Spoilage Sensitivity
- Columns – Year
- Values – Revenue
- Filters – Region (Northeast) and Category (Dairy)

This Dice pivot table displays a subset of data based on multiple conditions enabling multidimensional analysis of combining several criteria.

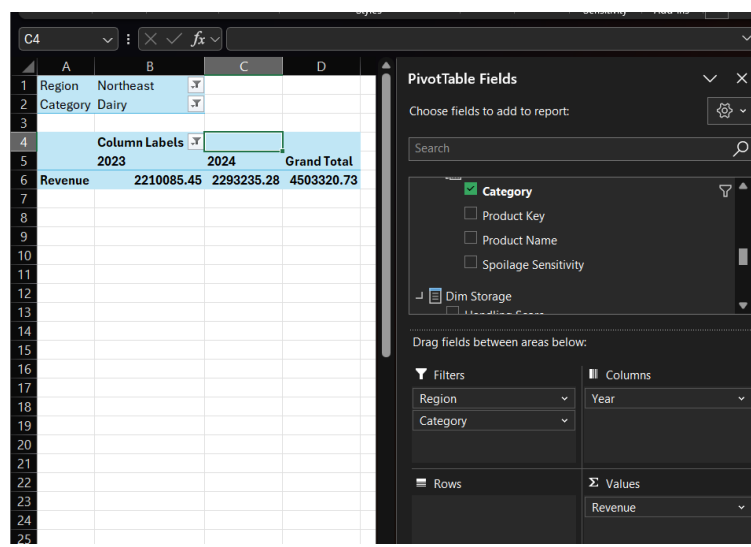


Figure 38: Dice Operation filtering multiple dimensions

3.5 Pivot (Rotate Data)

Pivot is a visualization operation which rotates the data axes to provide an alternative presentation of the data. It may contain swapping the rows and columns or moving one of the row-dimensions into the column dimensions. This demonstrated by changing the orientation of the Pivot Table.

Example Configuration 1:

- Rows – Spoilage Sensitivity
- Columns – Year
- Values – Revenue

Example Configuration 2:

- Rows – Year
- Columns – Spoilage Sensitivity
- Values – Revenue

These 2 configurations result the same data view from different perspectives by swapping rows and columns to help users to analyse relationships between dimensions more efficiently.

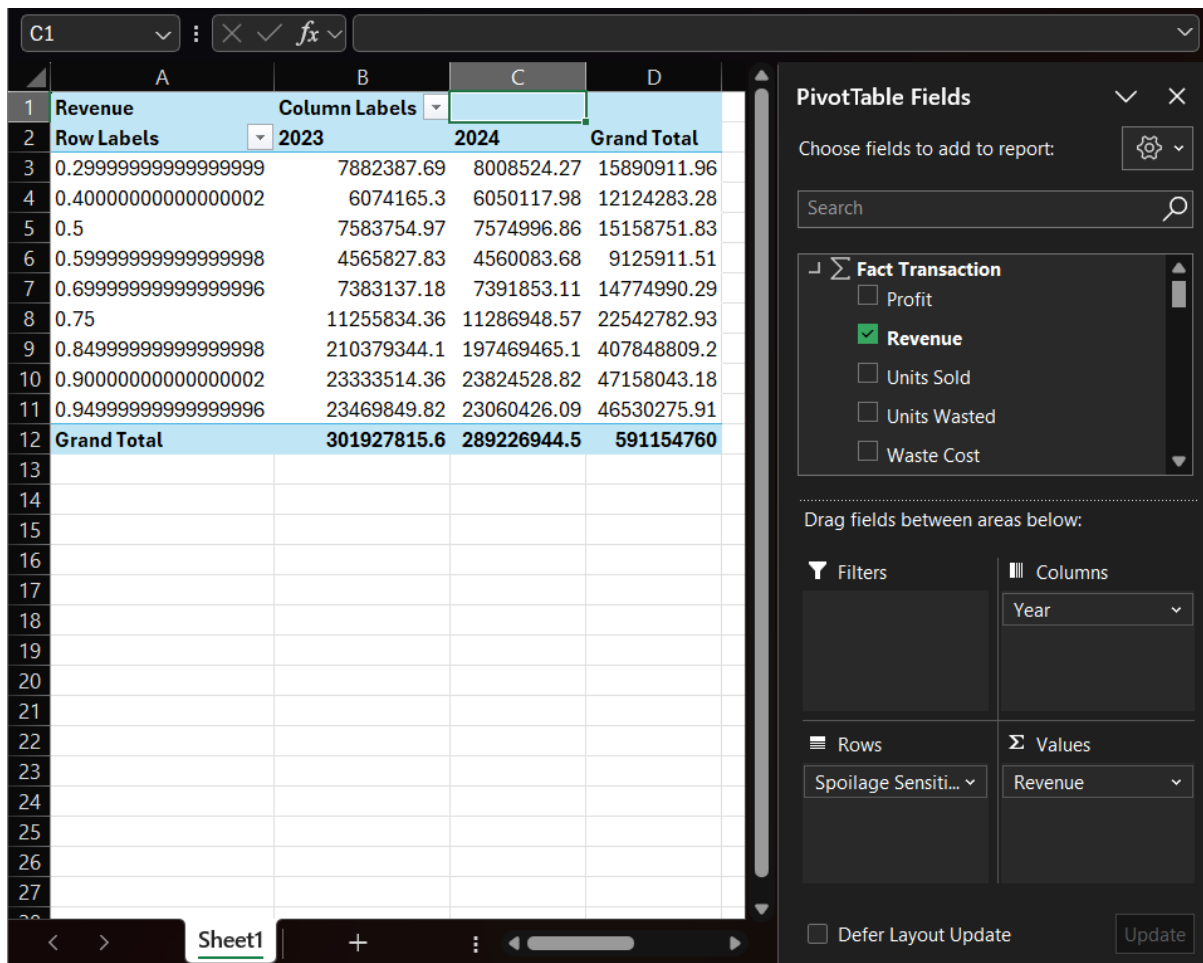


Figure 39: Pivot Configuration 1 (Spoilage Sensitivity by Year)

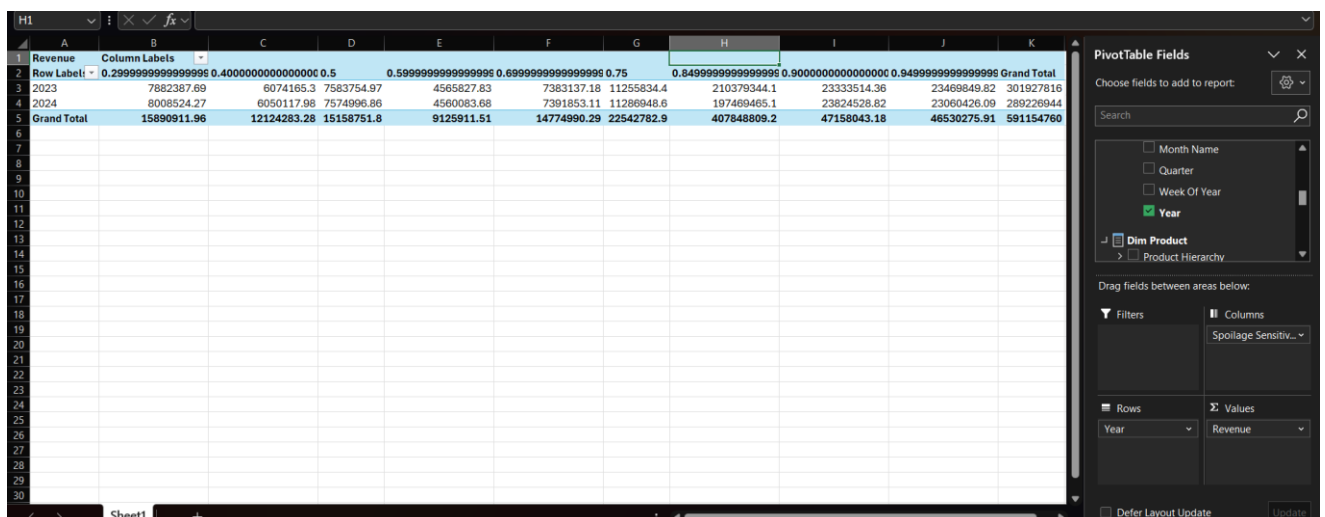


Figure 40: Pivot Configuration 2 (Spoilage Sensitivity by Year)

4. Visualization using Pivot Column Charts

Pivot Charts were created based on Pivot Tables and use to enhance data interpretation and visualization. The Pivot Chart, axis represents the categorical dimension while the Values represent numerical measure. The Pivot Charts visualizations improve understanding of patterns, comparisons, and trends.

Chart 1:

- Axis – Year
- Values – Revenue
- Purpose – this chart shows yearly revenue trends.

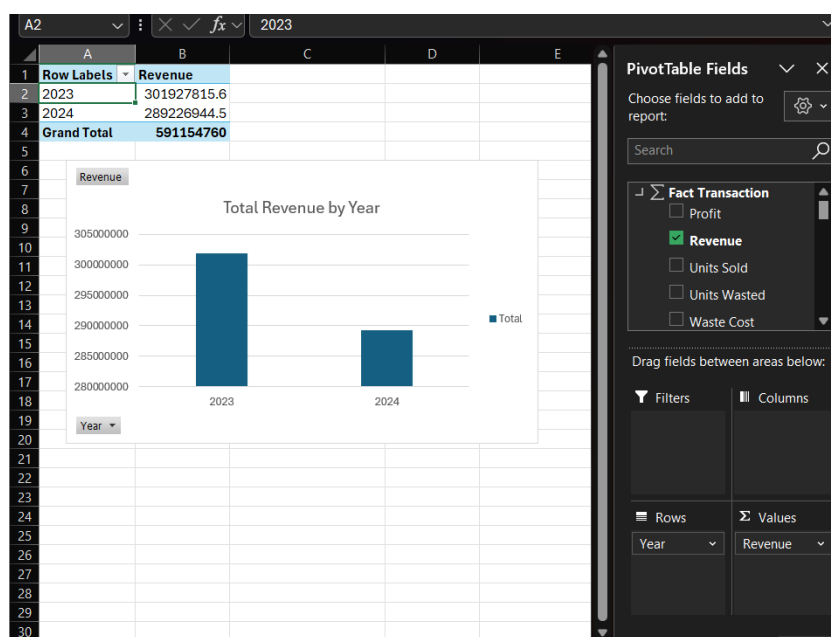


Figure 41: Year Revenue Chart

Chart 2:

- Axis – Spoilage Sensitivity
- Values – Revenue
- Columns - Year
- Purpose – this chart compares revenue across spoilage sensitivity categories over time.

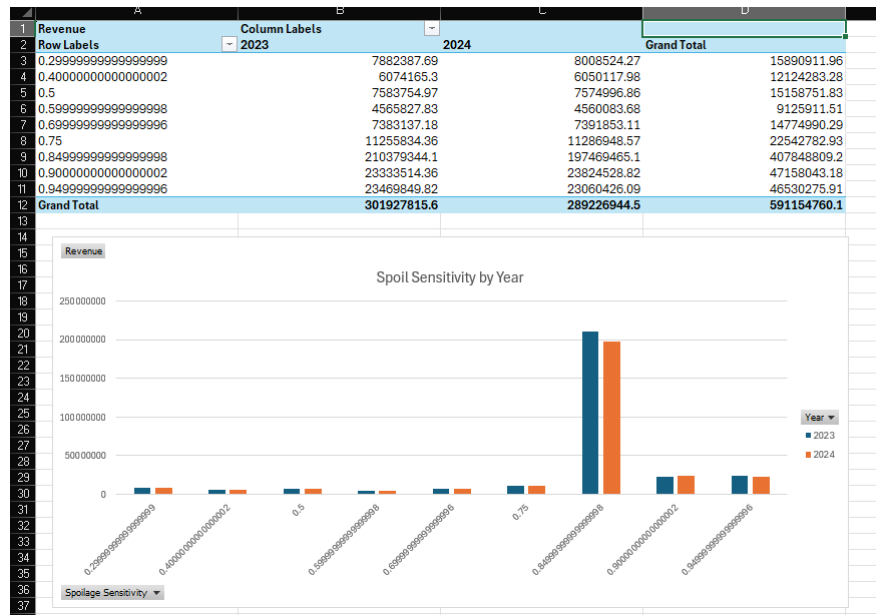


Figure 42: Spoilage Sensitivity by Year

Chart 3 (Pivot View):

- Axis – Year
- Values – Revenue
- Columns - Spoilage Sensitivity
- Purpose – this chart provides an alternative perspective of the same data by rotating data axes.

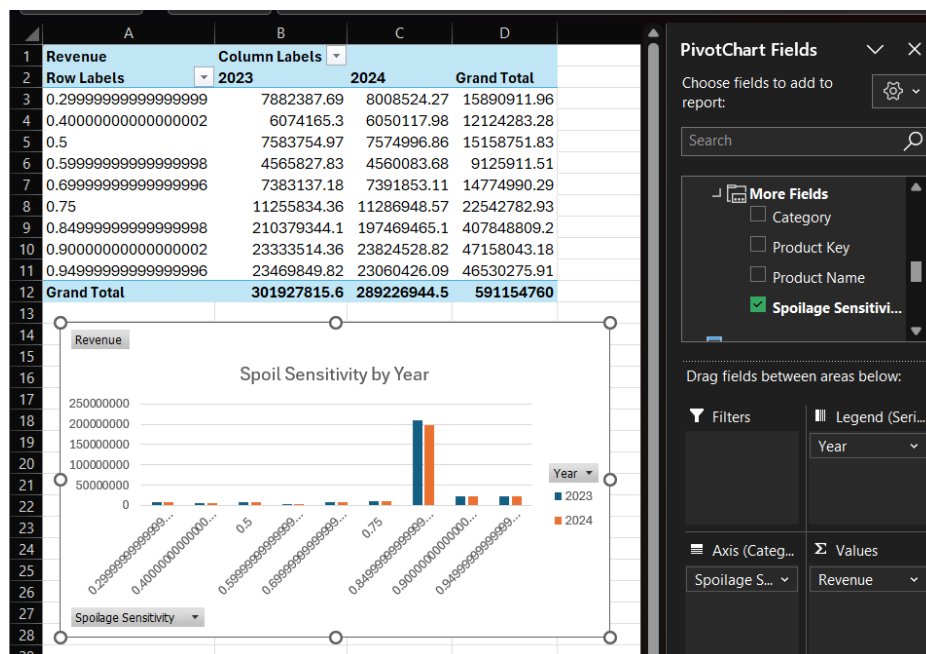


Figure 43: Spoilage Sensitivity Over Time

Step 4: Power BI Reports

1. Overview and Importance of Power BI Reports

The Power BI reports are designed to change the nature and content of the information stored in the Perishable_SSAS SQL Server Analysis Services into a visual and interactive format. The reports are important for business analysis and decision-making. The importance of the Power BI reports is evident through the following:

- Data conversion into a visual and understandable format
- Interactive analysis through filters, slicers, and drill functionality
- Interactive analysis through real-time exploration of business metrics such as revenue, profit, and waste
- Decision-making for business areas such as inventory, suppliers, and waste

Power BI reports are a powerful tool for business intelligence that bridges the gap between business data and meaningful insights.

2. Data Preparation

The reports are designed to use the Perishable_SSAS SQL Server Analysis Services as a source for business analysis and decision-making. The data preparation step was performed to:

- Load the Perishable_SSAS Cube into the Power BI tool.
- Verify that the data is of the appropriate type, such as numerical, categorical, or date
- Use the Dim_Date table for analysis Date Hierarchy (Year→Quarter→ Month_Name → Day)

The step was important for ensuring that the analysis is performed on a consistent and clean dataset.

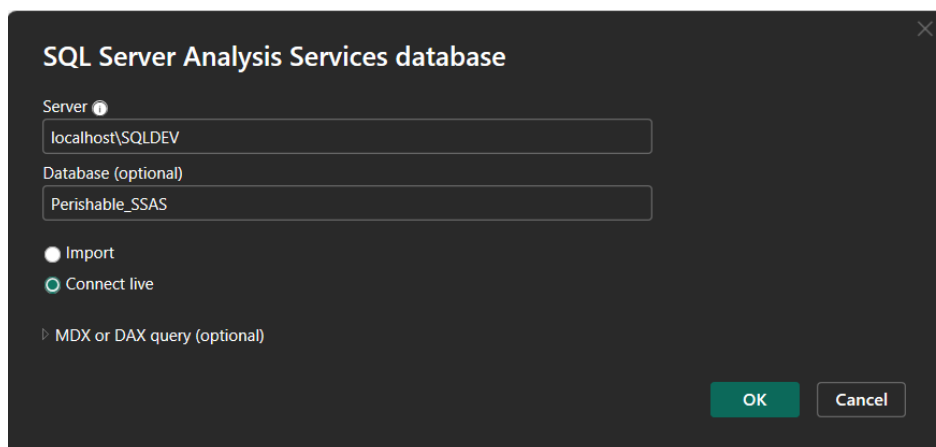


Figure 44: Data Warehouse connection

3. Data Modeling

The Power BI tool uses an SSAS Cube for business analysis and decision-making, where:

- The Fact_Transaction table includes measures such as revenue, profit, units sold, and waste cost
- The dimension tables, such as Dim_Product, Dim_Store, Dim_Supplier, Dim_Date, and Dim_Storage, are used for analysis

The tables are related to one another through surrogate keys. The SSAS Cube is important for improving query performance and analysis through multiple dimensions.

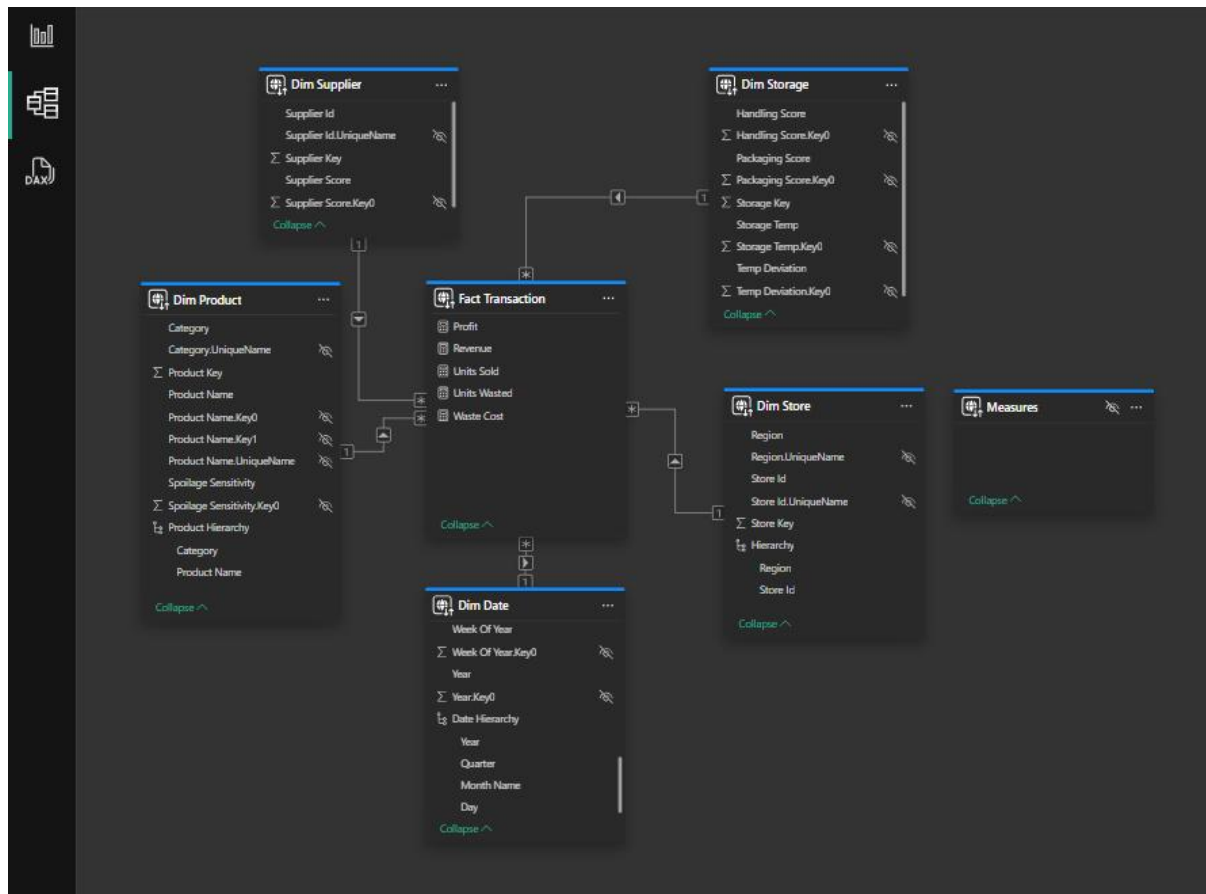


Figure 45: Model View SSAS Cube

4. Dashboards

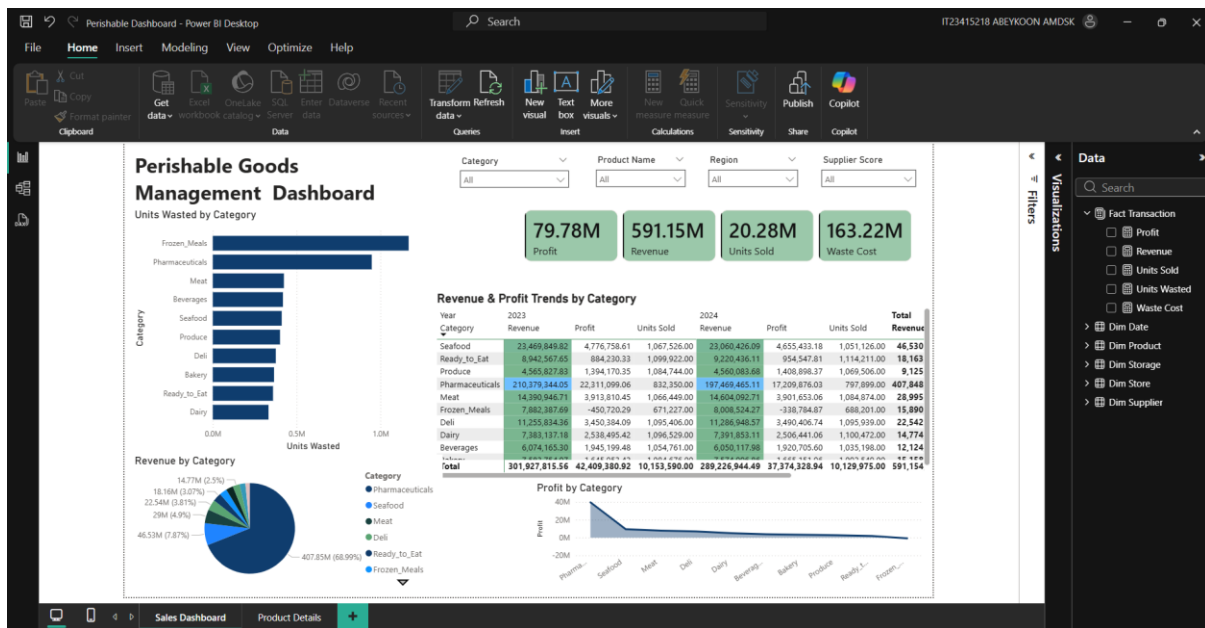


Figure 46: Sales Dashboard with Matrix Visual, Slicers and Cascading Filters

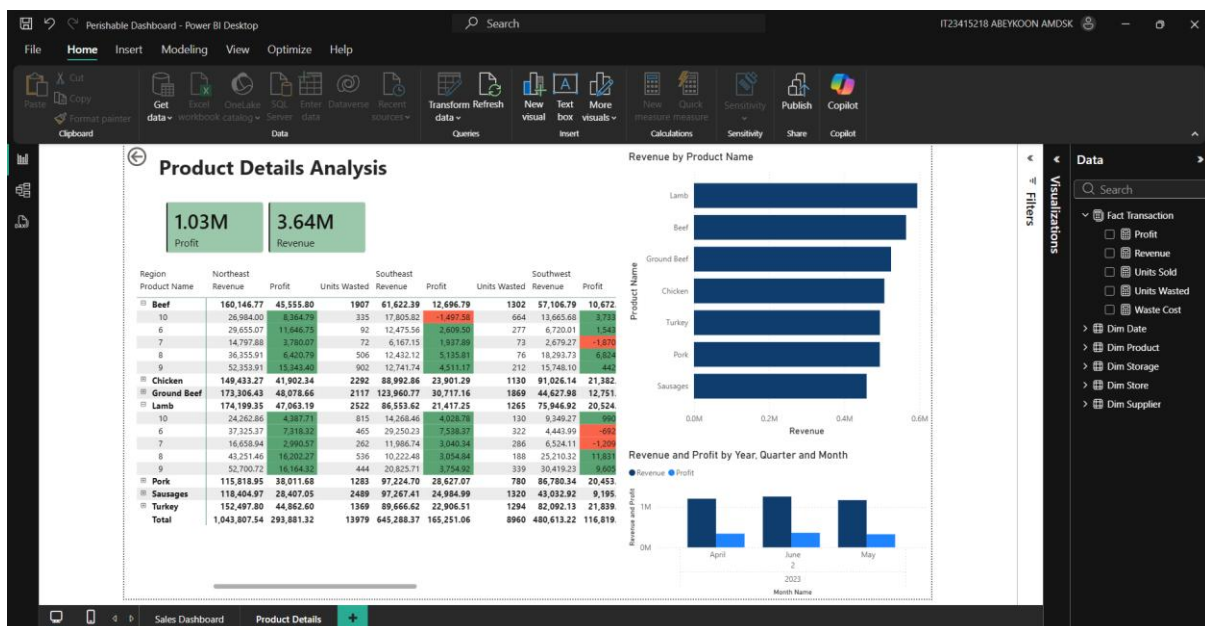


Figure 47: Product Details Analysis with Drill-Down (Date Hierarchy) and Drill-Through

Report 1: Matrix Visual and Performance Analysis

For Report 01, the Matrix visual was utilized to analyze the performance of product categories over the years.

- For the rows, the field Category was chosen, while the columns were based on the Year field.
- For the values, the fields Revenue, Profit, and Units_sold were selected.
- Conditional formatting was applied to highlight high and low values, enabling quick identification of best performance and underperformance categories.

This report provides an orderly comparison of the performance of product categories, allowing users to determine which categories are the best or worst performers.

Revenue & Profit Trends by Category

Year	2023			2024			Total
Category	Revenue	Profit	Units Sold	Revenue	Profit	Units Sold	Revenue
Seafood	23,469,849.82	4,776,758.61	1,067,526.00	23,060,426.09	4,655,433.18	1,051,126.00	46,530
Ready_to_Eat	8,942,567.65	884,230.33	1,099,922.00	9,220,436.11	954,547.81	1,114,211.00	18,163
Produce	4,565,827.83	1,394,170.35	1,084,744.00	4,560,083.68	1,408,898.37	1,069,506.00	9,125
Pharmaceuticals	210,379,344.05	22,311,099.06	832,350.00	197,469,465.11	17,209,876.03	797,899.00	407,848
Meat	14,390,946.71	3,913,810.45	1,066,449.00	14,604,092.71	3,901,653.06	1,084,874.00	28,995
Frozen_Meals	7,882,387.69	-450,720.29	671,227.00	8,008,524.27	-338,784.87	688,201.00	15,890
Deli	11,255,834.36	3,450,384.09	1,095,406.00	11,286,948.57	3,490,406.74	1,095,939.00	22,542
Dairy	7,383,137.18	2,538,495.42	1,096,529.00	7,391,853.11	2,506,441.06	1,100,472.00	14,774
Beverages	6,074,165.30	1,945,199.48	1,054,761.00	6,050,117.98	1,920,705.60	1,035,198.00	12,124
Total	301,927,815.56	42,409,380.92	10,153,590.00	289,226,944.49	37,374,328.94	10,129,975.00	591,154

Figure 48: Report 1 - Matrix visual design

Report 2: Interactive Analysis with Slicers and Cascading Filtering

For Report 02, created an interactive analysis using several slicers.

- The fields Category, Product_Name, Region, and Supplier_score were chosen as the fields in the slicers.
- Cascading filtering was implemented, where the other slicers would dynamically filter when one of the slicers is selected.
- KPI cards were created for the Total Revenue, Total Profit, Total Waste_Cost and Total Units_Sold.
- Line chart was created to analyze the total profit by category,
- Bar/column chart was implemented to identify and analyze total units_wasted by category.
- Pie Chart represents total revenue by category.

This report allows users to interactively analyze the data based on the factors selected, including product categories, region, and supplier performance.

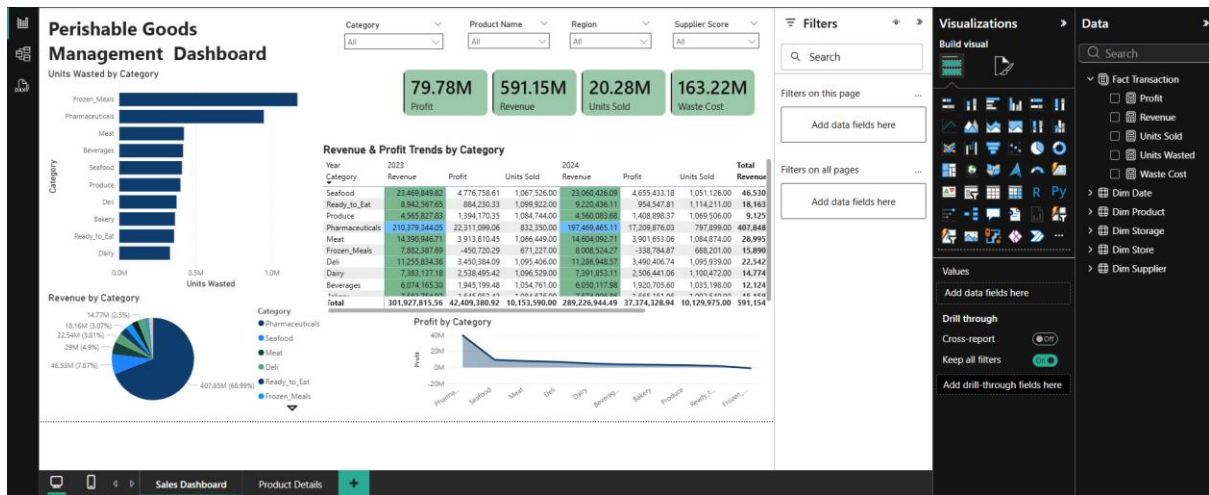


Figure49: Report 2 - Slicers & Cascading Filters

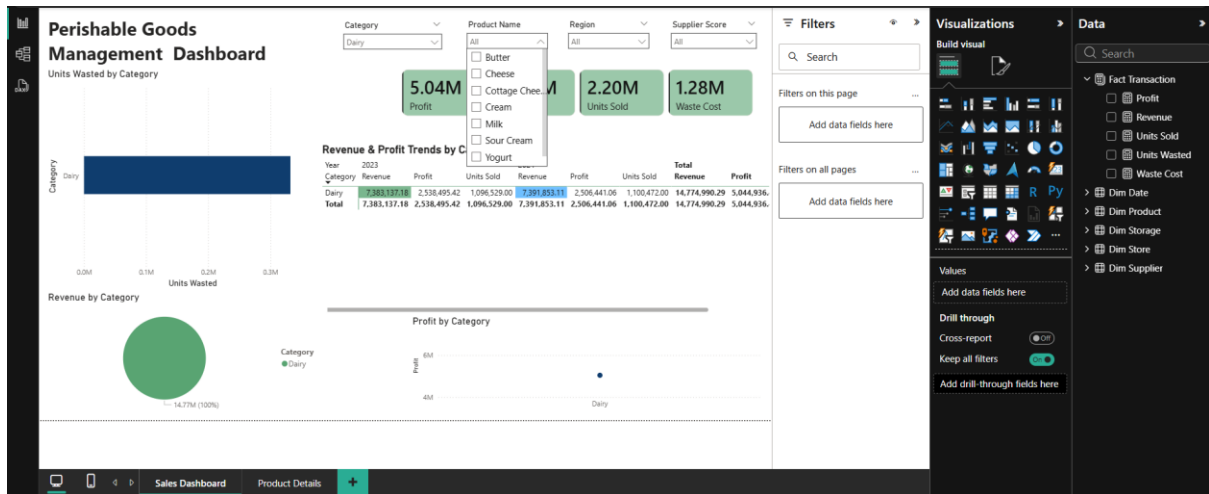


Figure 50: Report 2 - Cascading applied to Dairy Product Category

Report 3 – Drill-Down Hierarchical Analysis

For Report 03 includes a drill-down chart to support hierarchical data exploration.

- For the hierarchy, the fields Year, Quarter, and Month, Day were selected.
- Bar Chart was created to allow users to drill down

This report allows users to navigate through different levels of time to analyze the data in a hierarchical manner, helping determine the seasonal variance and detailed performance insights.

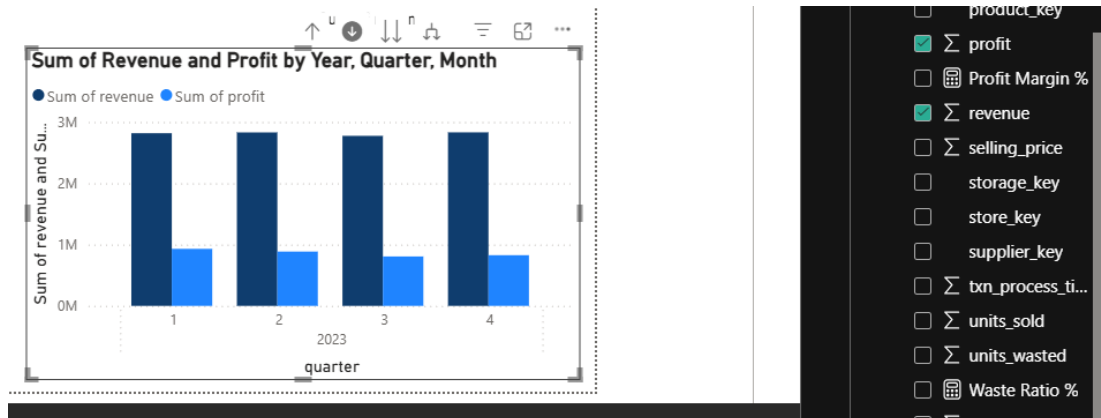


Figure 51: Drill-Down to Year 2023 Quarter Vise Revenue Analysis

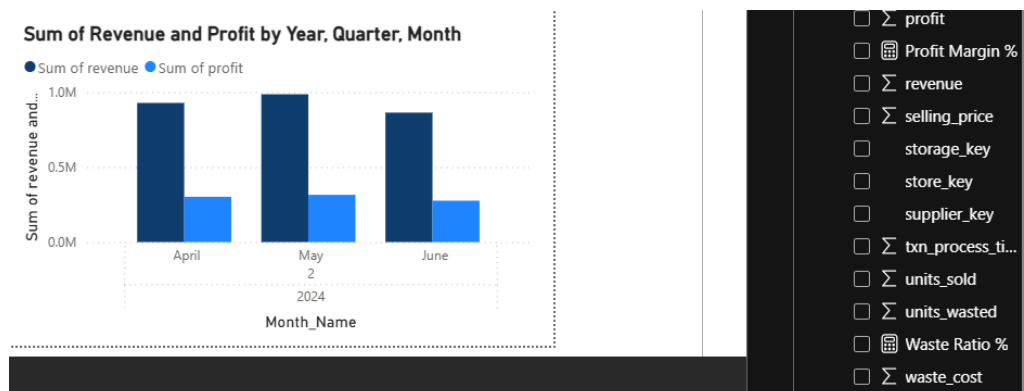


Figure 52: Report 3 - Drill-Down to year 2024, Quarter 2 Months

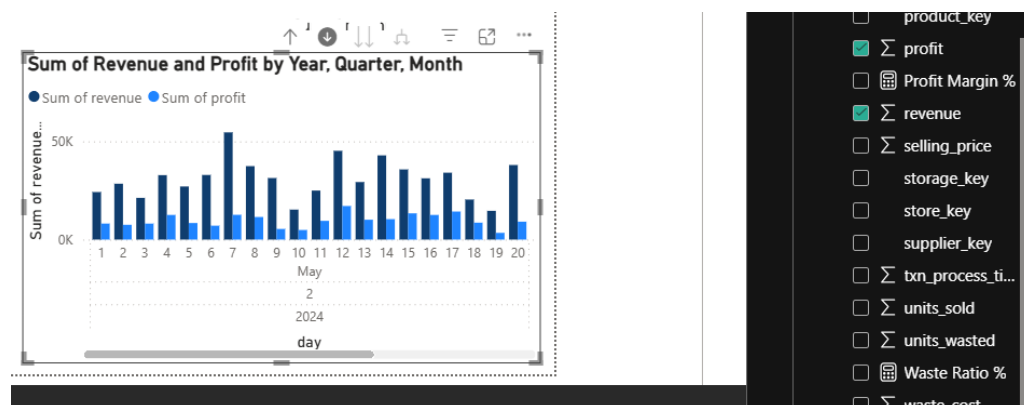


Figure 53: Drill-Down to Year 2024, Quarter 2, May Daily Revenue Analysis

Report 4: Drill-Through Product Details Analysis

Report 04 included the implementation of a drill-through feature to enable in-depth analysis.

- Summary visuals were created to display the main information such as product details matrix, a stacked bar chart for total revenue by product.
- A separate detail page (Product Details) was designed to display the product details, with 'Category' being the filter used for drill-through.
- Supplier_Score attribute represents that higher supplier scores are associated with better product performance and lower waste. This helps to identify lower performance suppliers.
- Users can right-click on the category to navigate to the details page.
- 2 different KPIs were implemented to identify the total profit and revenue.
- The Product details matrix highlights both profitable and loss products sales that negative profit values help identify operational inefficiencies like high waste or poor supplier performance.
- Conditional formatting was implemented to highlight negative and positive profit values for better visual analysis.

This report includes in-depth analysis using the drill-through feature to provide more information for the users.

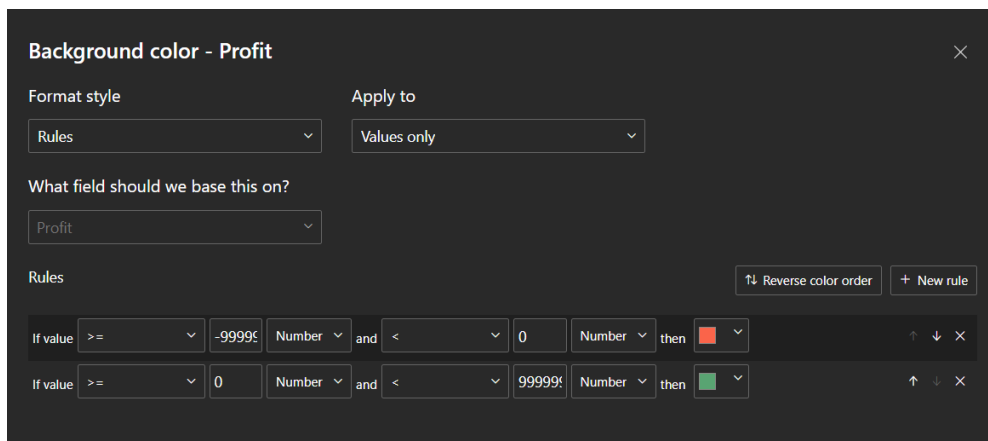


Figure 54: Conditional Formatting for Total Profit

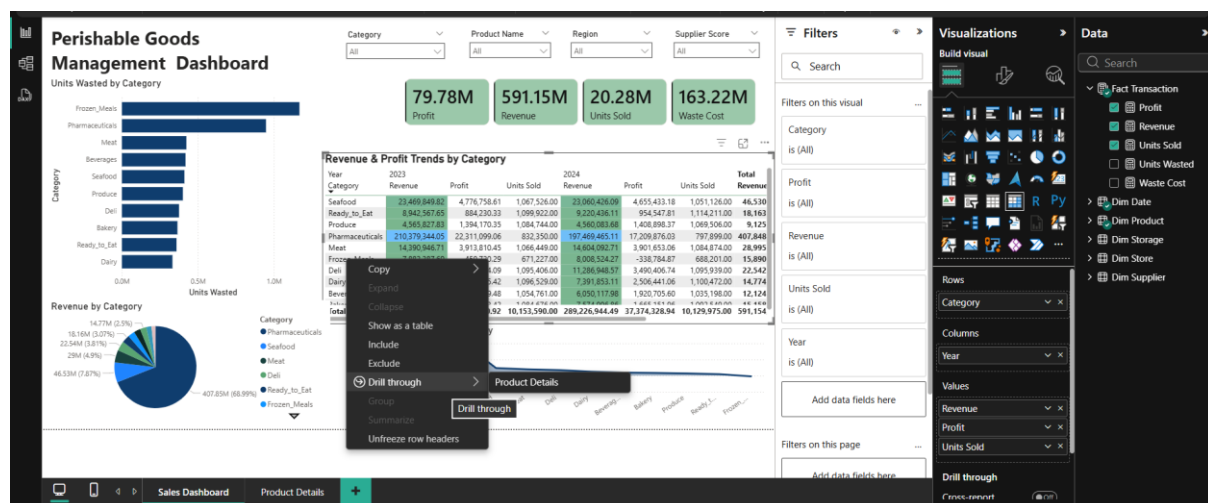


Figure 55: Drill-Through Ready-to eat Product Category



Figure 56: Ready to eat Products Details Analysis

5. Visual Design

A consistent and user-friendly visual design was used for each of the reports:

- Titles and labels are clear for each visual
- Logical flow of the visualizations, starting with KPIs, followed by trends, and finally analysis
- Consistent use of colors according to the domain of ‘perishable goods’
- ‘Green’ is used to represent KPI cards
- ‘Dark Blue’ is used to represent Chart Columns
- ‘Neutral colors’ are used to represent ‘Background’
- ‘Red’ is used to represent Loss Profits.

6. Publish the Dashboard Reports to Power BI

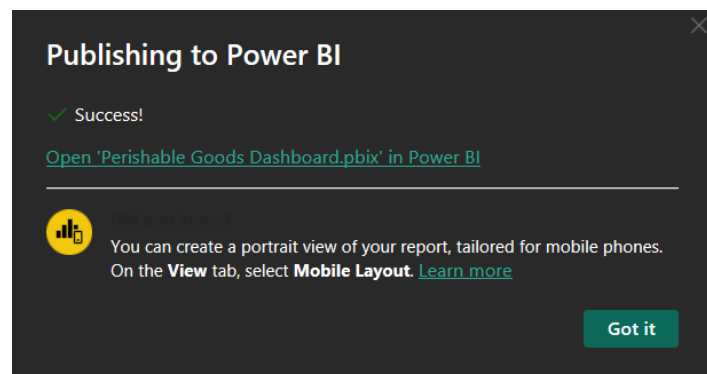


Figure 57: Publish reports